
LADDER: LLM Agent Diagnostic of Domain-level Estimation Reliability

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Users learn where each LLM agent is reliable; the agent itself does not. We ask
2 a single mechanistic question: at which sub-capability does this self-modeling
3 break down? Three candidates—H1 output channel, H2 prior metacognition,
4 H3 noise-robust online inference—are conflated by standard benchmarks. We
5 build **LADDER**, a diagnostic benchmark whose five-level oracle ladder (L0 raw
6 confidence $\rightarrow$ L4 full-statistic oracle) lets each adjacent comparison rule out one
7 candidate, scored with **Capability Boundary Fidelity (CBF)**, a per-domain metric
8 distinct from per-query calibration (ECE). The ladder yields a clean localization
9 across 15 models. H1 is ruled out (GT-SelfStats CBF ≥ 0.99 on every model);
10 H2 is partial and family-split rather than the bottleneck (GPT models reach 0.795–
11 0.859 zero-shot, while the three Claude models fall below NoMemory). H3 is
12 the locus: first-order single-rater learners cluster near NoMemory, while explicit
13 *second-order* estimators (asymmetric Dawid–Skene, GLAD) substantially exceed
14 the L1 prior, with GLAD winning on all six frontier models (binomial sign-test
15 $p = 0.016$). The pattern replicates across task families (12/15 sign-flip rate on
16 extension tasks), implying that per-task boundary modeling is also needed. We
17 name this failure mode **Self-Boundary Collapse** and identify noise-robust second-
18 order online inference as the central remaining challenge. We release the ladder,
19 CBF, cross-family user simulator, and trace dataset as benchmark infrastructure.

20 1 Introduction

21 LLM agents are increasingly used in long-horizon, multi-turn settings, including customer service,
22 coding, research, and tool use. Over time, users develop an empirical sense of where an agent is
23 reliable. The agent, however, typically retains no comparable self-model: each session begins without
24 prior calibration and ends by discarding its own evidence. This asymmetry is well documented, but
25 the underlying *capability gap* remains poorly localized. In particular, we do not yet know *where*,
26 mechanistically, agent self-modeling fails.

27 We distinguish three possible bottlenecks. Under H1, the output channel is the limiting component:
28 the model cannot format and emit numerical per-domain statistics. Under H2, the model lacks an
29 accurate prior over its own per-domain abilities, so any useful self-assessment must be *learned*
30 through interaction. Under H3, the model can state reasonable priors but cannot update them
31 reliably from *noisy* implicit user feedback. These hypotheses imply different intervention targets.
32 H1 suggests fine-tuning for numerical reporting; H2 points to pre-training-time metacognitive
33 curricula; H3 calls for noise-robust online estimation. Existing work on agent self-knowledge [3, 16],
34 calibration [13, 18, 45], and memory systems [26, 33, 35, 41, 47] measures important symptoms but
35 does not isolate which of these bottlenecks is responsible. We refer to the resulting failure mode,
36 in which LLM agents repeatedly receive feedback yet fail to convert it into a stable domain-level

self-model, as **Self-Boundary Collapse**. Our central question is where this failure appears on the H1/H2/H3 ladder.

We study this question with LADDER. The benchmark is organized around an oracle-bracketed *five-level baseline ladder*: L0 raw confidence (NoMemory), L1 zero-shot per-domain prior, L2 noisy-feedback learners (LLM-self-assessment, classical recalibration, EM-joint, second-order Dawid-Skene/GLAD), L3 binary-label oracle, and L4 full-statistic oracle. Adjacent comparisons localize the failure. L4–L3 tests the output channel (H1), L3–L1 tests the prior (H2), and L3–L2 tests online inference from noisy feedback (H3). The protocol pairs each agent with a domain-heterogeneous simulated user, reliable 75% of the time in-expertise and 46% out, with cross-family routing to mitigate a dual-role bias that we quantify in §4.3. Sessions last 50 turns, and the resulting per-domain self-model is evaluated with **Capability Boundary Fidelity (CBF)**, a metric we introduce to complement per-query calibration measures such as ECE.

Across 15 models from 7 families and 47K+ interactions, the ladder gives a clear localization. H1 is ruled out: GT-SelfStats reaches $\text{CBF} \geq 0.99$ on every model, showing that the output channel can faithfully emit per-domain statistics when supplied. H2 is real but not the main bottleneck. The zero-shot prior is the strongest non-oracle baseline for GPT models, with CBF 0.795 on GPT-5.4 and 0.859 on GPT-5.5, but it falls below NoMemory on all three Claude models, 0.460–0.482 vs. 0.557–0.603; the frontier mean is 0.640. Thus prior self-knowledge exists in some families and is absent in others, but it never reaches the oracle ceiling. The main failure is H3. First-order single-rater learners, including UF-PDC, Platt, HistBin, Bayes, and simple EM-Joint, cluster near NoMemory on the frontier, with CBF 0.628–0.634, and cannot recover per-domain accuracy from noisy feedback. Explicit *second-order* estimators do substantially better: asymmetric Dawid-Skene reaches a frontier mean of 0.755, and GLAD reaches 0.712. GLAD beats L1 on all six frontier models, with Wilcoxon two-sided $p = 0.031$. A remaining L2→L3 gap of roughly 0.25 CBF on the frontier (best L2 DS-asym 0.755 vs. OraclePDC 1.000) persists, but it appears tractable rather than fundamental.

2 Related Work

LLM self-knowledge and calibration benchmarks. A growing line of work probes whether LLMs can represent or report their own correctness on a single query. Internal-representation studies show that latent confidence signals exist [4, 17, 18], while recent benchmarks evaluate metacognition behaviorally and find that frontier models exhibit only partial, context-dependent self-monitoring [1, 30]. Table 1 positions the most closely related benchmarks against LADDER along four design axes: feedback channel, horizon, whether self-knowledge is accumulated, and whether the protocol forces second-order user-reliability inference.

Table 1: Related-benchmark positioning along four design axes. LADDER is the only benchmark with a noisy interactive feedback channel that the agent must accumulate into a per-domain self-model under second-order user-reliability uncertainty.

Benchmark	Feedback	Horizon	Self-knowledge	2nd-order
SAD [22], SelfAware [51]	none	single-shot	static probe	—
Barkan [3]	clean step signals	multi-step exec	per-task	—
MemoryAgentBench [16]	implicit (mem ops)	long-horizon	exogenous mem only	—
AgentBoard [31], τ -bench [49]	task reward only	multi-turn	none	—
Knowledge-boundary [24, 44]	none	single-shot	static	—
LADDER (ours)	noisy NL, 75/46 asym	50-turn session	per-domain accumulated	✓

(i) *Single-shot self-knowledge probing.* SAD [22] and SelfAware [51] score self-knowledge per query without temporal dynamics; their findings supply the “L1 prior” anchor of our ladder. LADDER uses the L1→L3 contrast to ask whether *noisy interaction* improves on that prior—finding it does not.

(ii) *Multi-step agent overconfidence.* Barkan et al. [3] measure whether 13 frontier LLMs can predict their own success on BigCodeBench and SWE-Bench under multi-step tool execution and find all models overconfident with overconfidence *worsening* during execution. Their setting gives clean per-step success/failure signals; ours gives noisy domain-dependent natural-language feedback, where first-order methods cluster at the L2 floor. Barkan establishes that overconfidence persists

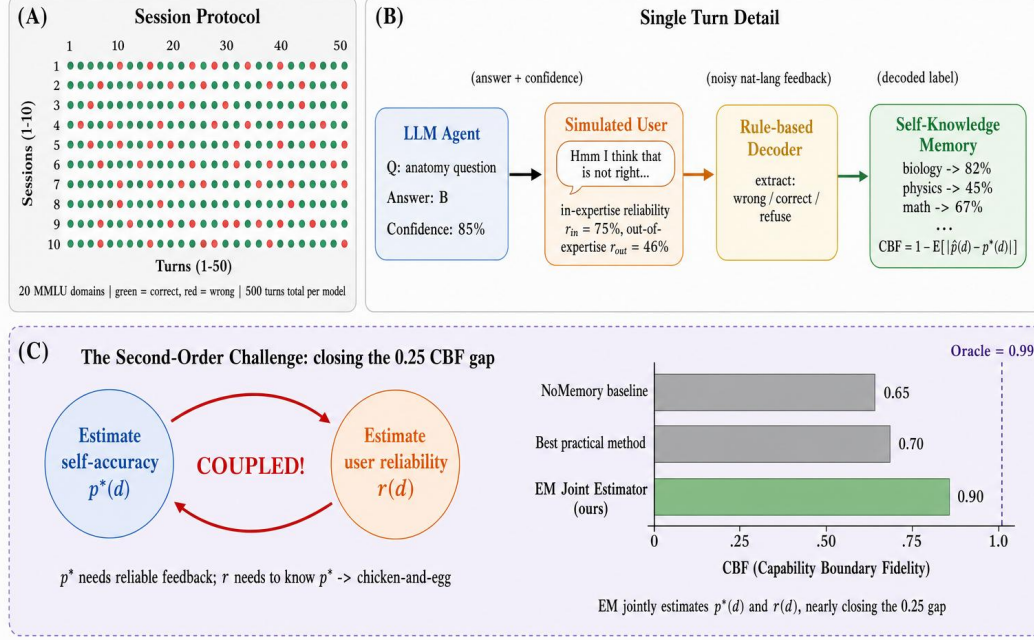


Figure 1: **The L0–L4 baseline ladder localizes where agent self-knowledge fails.** L4≈L3≈1.0 rules out the output channel (H1); the L2→L3 gap of ~0.30 isolates noise-robust online inference (H3) as the bottleneck. **(A)** Session protocol: 50 sessions × 50 turns over 20 MMLU domains (green=correct, red=wrong). **(B)** Single turn: the agent answers with confidence; a cross-family simulated user gives domain-dependent noisy feedback ($r_{in} \approx 75\%$ in-expertise, $r_{out} \approx 46\%$ out); a decoder extracts the labeled signal that updates the per-domain self-model. **(C)** The five-level diagnostic ladder: L0 raw confidence (NoMemory) → L1 zero-shot prior (ZS-SK) → L2 noisy-feedback learners (LSA, EM-Joint, DS-asym, GLAD) → L3 binary-label oracle (OraclePDC) → L4 full-statistic oracle (GT-SelfStats). Adjacent comparisons isolate each candidate failure: L4–L3 isolates H1, L3–L1 isolates H2, L3–L2 isolates H3.

78 under clean traces; we localize *where* the failure lives when feedback itself is noisy, via the L0–L4
79 oracle-bracketing ladder.

80 (iii) *Multi-turn agent benchmarks without self-knowledge.* AgentBoard [31] and τ -bench [49] evaluate
81 task completion under multi-turn interaction but do not measure self-knowledge; their reward signal
82 is task success, not self-estimate accuracy. LADDER holds the underlying task simple (MCQ)
83 and varies only the information-access regime, so the diagnostic ladder isolates self-knowledge
84 mechanisms these benchmarks cannot disentangle from task-execution confounds.

85 Static knowledge-boundary benchmarks [12, 15, 24, 27, 28, 44, 50] share the same structural lim-
86 itation: they evaluate *a priori* boundary awareness, not whether the agent’s self-model can be
87 accumulated under noisy interactive turns (Appendix R).

88 **Agent memory and learning from implicit feedback.** Existing memory systems—MemGPT [35],
89 Mem0 [33], RMM [41], A-MEM [47], MemOS [26], MEM1 [54]—and the recent MemoryAgent-
90 Bench [16] all store *exogenous* state (user facts, dialogue, trajectories). None model the agent’s *own*
91 per-domain reliability; we introduce this as a fifth memory competency. On the feedback-learning
92 side, RLHF [34] and DPO [37] update weights at training time assuming reliable annotators; Liu
93 et al. [29] establish that implicit dialogue feedback is noisy and only partially useful as a distillation
94 signal; Leng et al. [23] show RLHF itself induces systematic overconfidence. LADDER studies
95 the orthogonal regime: *inference-time* accumulation of *noisy, domain-dependent* feedback into a
96 per-domain self-model, with no parameter updates. The unique twist is asymmetric reliability (75%
97 in-expertise vs. 46% out), which turns boundary learning into a *second-order* problem.

Selective prediction and task-specific calibration. Per-query selective-prediction methods [2, 9, 19, 20, 25, 42, 46] treat each prediction in isolation. LADDER instead enables *historical* selective prediction: abstention based on persistent per-domain self-model statistics. Independently, prior work has noted that calibration does not transfer across tasks [6, 32, 39]; our experiments quantify this systematically across MMLU, GSM8K, HumanEval, and BFCL. The closest concurrent benchmarks [5, 10, 11, 14, 53] either assume perfect feedback, require weight updates, or focus on single-agent dynamics; none combine multi-task evaluation, noisy domain-dependent feedback, and a quantitative boundary-fidelity metric over extended horizons.

3 LADDER

LADDER evaluates a single capability: given a stream of (q_t, a_t, c_t, f_t) tuples over T turns—a question q_t from a hidden domain d_t , the agent’s answer a_t with confidence $c_t \in [0, 1]$, and a natural-language user feedback f_t —can the agent build a function $\hat{p} : \mathcal{D} \rightarrow [0, 1]$ that closely approximates its true per-domain accuracy $p^*(d)$?

The benchmark instantiates this question via a 50×50 protocol: 50 sessions $\times$ 50 turns per session, totaling 2,500 turns per model, drawing from 20 MMLU subjects [15] partitioned per model into Strong/Medium/Weak tiers relative to that model’s own accuracy distribution. The user is simulated by a cross-family LLM (avoiding the dual-role bias of Appendix M), parameterised by an expertise set $\mathcal{E} \subset \mathcal{D}$ with empirically calibrated reliabilities of $\sim 75\%$ in-expertise and $\sim 46\%$ out.

We additionally construct three extension tasks—GSM8K [8], HumanEval [7], and BFCL tool-calling [48], collectively called the *extension suite*—to test cross-task generalisation (Appendix P). Full task formalisation, session protocol, question-pool partitioning, and persona parameterisation are deferred to Appendix B.

Capability Boundary Fidelity (CBF). Our central metric scores how closely the agent’s self-assessed per-domain accuracy tracks the truth:

$$\text{CBF} = 1 - \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} |\hat{p}(d) - p^*(d)|, \quad \text{CBF} \in [0, 1]. \quad (1)$$

CBF is computed once per session at the end of the 50-turn protocol. CBF is complementary to per-query calibration metrics (ECE/Brier): ECE operates on (c_t, y_t) pairs while CBF operates on $(\hat{p}(d), p^*(d))$ pairs, so neither implies the other; Figure 2 confirms this empirically (Pearson $r = +0.21$, $p = 0.45$ across 15 models). The trivial estimator $\hat{p}(d) = 0.5$ attains $\text{CBF} = 1 - \mathbb{E}_d[|0.5 - p^*(d)|]$, which is high for weak models clustered near 50% and low for strong models with widely varying accuracy across domains; this asymmetry is exactly what makes the metric informative for our diagnostic.

Baseline-adjusted CBF: the primary cross-model metric. To remove the trivial-estimator artifact when comparing methods *across* models, we define a normalized variant that measures the fraction of the oracle gap a method closes relative to the constant baseline:

$$\widetilde{\text{CBF}}(m, \text{method}) = \frac{\text{CBF}_{\text{method}}(m) - \text{CBF}_{\text{NoMem}}(m)}{\text{CBF}_{\text{Oracle}}(m) - \text{CBF}_{\text{NoMem}}(m)}, \quad (2)$$

where $\text{CBF}_{\text{NoMem}}(m)$ uses the model’s verbalized confidence (which empirically clusters near 0.5 on the trivial domains) as the constant-baseline anchor, and $\text{CBF}_{\text{Oracle}}(m) \geq 0.99$ is the L3 OraclePDC ceiling. Values are interpretable as “fraction of L0→L3 gap closed”: $\widetilde{\text{CBF}} = 1$ matches oracle, 0 ties the trivial baseline, < 0 underperforms it. We report CBF as the primary metric for cross-model means and ladder-headline comparisons (§4.2), and use raw CBF only as a within-model diagnostic.

Auxiliary metrics and domain structure. We additionally report Hallucination Rate, ECE, and Brier as auxiliary calibration metrics; their definitions follow Guo et al. [13] and are given in Appendix B. CBF requires a multi-domain task: for single-domain extension tasks (GSM8K, HumanEval, BFCL) we report only ECE and confidence–accuracy gap, and the cross-task analysis in Appendix P uses these standard metrics.

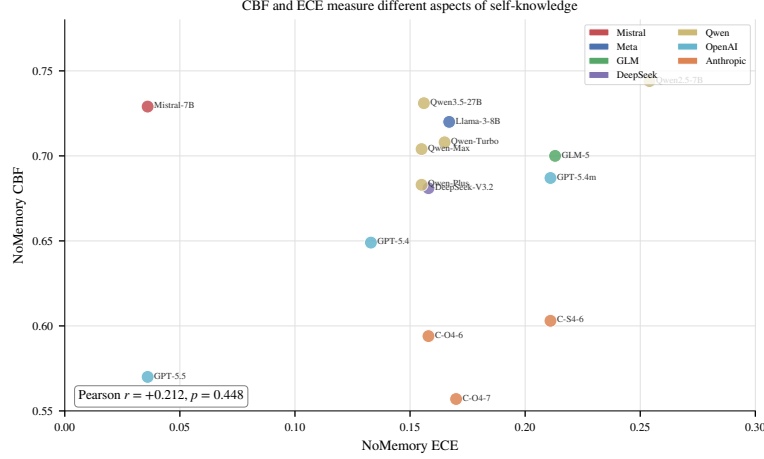


Figure 2: **CBF and ECE measure different aspects of self-knowledge.** Per-model NoMemory CBF vs. NoMemory ECE across all 15 models. Pearson $r = +0.21$ ($p = 0.45$) confirms the two metrics are essentially uncorrelated: a model can be well-calibrated per query (low ECE) and have poor per-domain self-knowledge (low CBF), or vice versa. CBF is therefore not a reparameterization of ECE.

Cross-model comparability. The 20 MMLU subjects are partitioned into Strong/Medium/Weak tiers *relative to each model’s own held-out validation accuracy* so that every model faces a non-trivial cross-domain spread. CBF is therefore a within-model diagnostic; cross-model means reported throughout (e.g., 0.685 for ZS-SK across all 15 models in Table 2) summarize per-model values that were each computed against the model’s own $p^*(d)$, and are appropriate for comparing *methods within the diagnostic ladder*, not for ranking models on absolute capability. We expand on this design choice and confirm headline conclusions are unchanged under an alternative absolute-cutoff partition in Appendix B.

Baselines: a five-level diagnostic ladder. We evaluate ten baselines organized as a ladder of *information access*, designed so that comparing adjacent levels isolates a specific sub-capability:

- **L0 (raw): NoMemory**—verbalized confidence with no memory, the stateless anchor.
- **L1 (prior only): ZS-SK** (ZeroShot Self-Knowledge)—the agent is asked, with no session history and no feedback, to estimate its own per-domain accuracy directly; this measures *prior metacognition*.
- **L2 (learning from noisy feedback): LSA** (LLMSelfAssess; in-context reasoning over session history), *UF-PDC* (LLM-decoded per-domain counting), *Platt-Online* [36], *HistBin-Online* [52], *Bayes-Domain* (Beta posterior per domain), and *EM-Joint* (joint estimation of $\hat{p}(d)$ and $\hat{r}(d)$). These differ in their inference machinery but all consume the same noisy feedback stream.
- **L3 (binary-label oracle): OraclePDC**—per-turn ground-truth correctness labels, with the agent computing per-domain running averages. Tests whether the agent could build a self-model *if feedback were clean*.
- **L4 (format-channel sanity ceiling): GT-SelfStats**—the agent is given $p^*(d)$ directly in the prompt and asked to emit them in the structured response format. This is a sanity check that the L2 gap cannot be attributed to an inability to format and emit per-domain statistics. We do not claim it tests verbalization-of-self-knowledge.

All feedback decoding (L2 + UF-PDC) uses an LLM-based decoder rather than rule-based pattern matching, more aligned with the “self-knowledge boundary” setting and avoiding rule-classifier coverage as a confound. Full per-baseline algorithms are in Appendix C.

Table 2: Main results: end-of-session CBF across 15 models on the five-level baseline ladder. Higher is better. **Bold** marks the best non-oracle (L0–L2) baseline per row. All 15 models use the **50 sessions** $\times$ **50 turns** protocol.

Model (Acc)	L0	L1	L2 (learn from noisy feedback)							Oracles	
	NoMem	ZS-SK	LSA	UF-PDC	Platt	HistBin	Bayes	EM	Best	OraPDC	GT-Stats
Mistral-7B (42%)	0.729	0.729	0.729	0.720	0.717	0.717	0.731	0.732	EM	1.000	1.000
Llama-3-8B (55%)	0.720	0.711	0.549	0.715	0.724	0.724	0.730	0.731	EM	1.000	1.000
GLM-5 (55%)	0.700	0.700	0.700	0.689	0.696	0.696	0.711	0.714	EM	1.000	1.000
DeepSeek-V3.2 (62%)	0.681	0.673	0.721	0.681	0.683	0.683	0.696	0.699	LSA	1.000	1.000
Qwen-Turbo (64%)	0.708	0.730	0.703	0.697	0.712	0.712	0.724	0.728	ZS	1.000	1.000
Qwen2.5-7B (65%)	0.744	0.663	0.701	0.738	0.745	0.745	0.755	0.757	EM	1.000	1.000
Qwen-Max (71%)	0.704	0.766	0.781	0.660	0.684	0.684	0.702	0.705	LSA	1.000	1.000
Qwen-Plus (79%)	0.683	0.735	0.435	0.686	0.695	0.695	0.700	0.701	ZS	1.000	1.000
Qwen3.5-27B (79%)	0.731	0.731	0.731	0.712	0.720	0.720	0.740	0.746	EM	1.000	1.000
<i>Frontier closed-source models:</i>											
GPT-5.4-mini (72%)	0.687	0.769	0.763	0.687	0.701	0.701	0.706	0.706	ZS	1.000	0.995
GPT-5.4 (82%)	0.649	0.795	0.781	0.668	0.675	0.675	0.674	0.677	ZS	1.000	0.994
GPT-5.5 (94%)	0.570	0.859	0.902	0.601	0.612	0.612	0.603	0.605	LSA	1.000	0.992
Claude-Opus-4-7 (69%)	0.557	0.482	0.641	0.593	0.577	0.577	0.571	0.570	LSA	1.000	0.992
Claude-Sonnet-4-6 (86%)	0.603	0.472	0.720	0.640	0.617	0.617	0.613	0.609	LSA	1.000	0.993
Claude-Opus-4-6 (90%)	0.594	0.460	0.680	0.618	0.597	0.597	0.603	0.604	LSA	1.000	0.993
<i>Within-tier means:</i>											
Frontier-v4 (n=6; all columns)	0.610	0.640	0.748	0.634	0.630	0.630	0.628	0.628	LSA	1.000	0.993
Strong-mid (n=3)	0.706	0.744	0.649	0.686	0.700	0.700	0.714	0.717	ZS	1.000	1.000
Mid (n=3)	0.711	0.689	0.708	0.705	0.713	0.713	0.725	0.728	EM	1.000	1.000
Weak (n=3)	0.716	0.713	0.659	0.708	0.712	0.712	0.724	0.726	EM	1.000	1.000

4 Diagnostic Results: Where Does Agent Self-Knowledge Fail?

4.1 Setup

We evaluate fifteen models from seven families spanning 42–90% MMLU accuracy (Appendix S). Each model–baseline pair runs 2,500 turns (50 sessions $\times$ 50 turns under the 50 $\times$ 50 protocol); user personas are simulated by a cross-family LLM with three expertise domains¹; answer extraction uses a lenient multi-strategy parser tolerant of free-form responses to avoid truncating reasoning models. Feedback decoding for all L2 baselines uses an LLM-based decoder. Open-weights models (Mistral-7B, Llama-3-8B) are served via vLLM from pinned snapshots and serve as the reproducibility anchor; API access dates, simulator-routing rules, and the parser specification are deferred to Appendix S.

4.2 Main Results

Three diagnostic patterns dominate Table 2: (i) algorithmic L2 methods collapse to NoMem on every frontier model; (ii) the zero-shot prior splits cleanly by family, with GPT estimates well-calibrated and Claude estimates inflated; (iii) only second-order estimators (DS-asym, GLAD) and in-context LSA exceed the L1 prior on the frontier. We examine each pattern below.

Algorithmic L2 collapses on frontier models; ZS and LSA do not. For both newly added frontier models, the five *algorithmic* L2 baselines (UF-PDC, Platt, HistBin, Bayes, EM-Joint) cluster near NoMem (GPT-5.5: 0.601–0.612, all $\sim$ 0.03–0.04 above NoMem 0.570; Claude-Opus-4-7: 0.570–0.593, all $\sim$ 0.01–0.04 above NoMem 0.557). With the 50 $\times$ 50 ZS/LSA values available, both LSA and the L1 prior break this collapse decisively on GPT-5.5 (LSA 0.902, ZS-SK 0.859: LSA is +0.290 over best algorithmic L2 0.612, ZS is +0.247 over the same) and on Claude-Opus-4-7 (LSA 0.641 vs. NoMem 0.557, +0.084). The pattern under the 50 $\times$ 50 protocol: algorithmic L2 methods fail

¹Frontier-set scope: 6 model endpoints (3 GPT, 3 Claude): GPT-5.4, GPT-5.4-mini, GPT-5.5, Claude-Opus-4-6, Claude-Sonnet-4-6, Claude-Opus-4-7. Main-table within-tier means, MMLU-Pro and BBH contamination controls (Appendix E, E), and Dawid–Skene/GLAD cross-model analysis all use this $n=6$ scope. $n=6$ BBH and MMLU-Pro CBF means are 0.811 and 0.849 (vs. MMLU ZS 0.640).

to surpass NoMem on every frontier agent; LSA cleanly exceeds NoMem on all 6 frontier rows; ZS exceeds NoMem on all 3 GPT rows but falls below NoMem on all 3 Claude rows. In particular, GPT-5.5 is now decisively dominated by LSA (0.902, the benchmark’s highest non-oracle CBF) followed by ZS (0.859); both substantially exceed every algorithmic L2 method (≤ 0.612) and the earlier 10×50 result in which LSA fell well below NoMem (0.623) does not reproduce. Held-out evaluation and second-order estimators (Dawid–Skene, GLAD) are reported on the 4 original frontier models; the MMLU-Pro and BBH contamination controls (Appendix E, E) cover all 6 frontier models.

H1 ruled out as the limiting factor ($L4 \approx 1.0$). GT-SelfStats reaches $CBF \geq 0.992$ on every one of the 15 models (worst gap to perfection 0.008 on Claude-Sonnet-4-6). When the true per-domain statistics are placed in the prompt, every tested model can format and emit them: the L2–L1 gap is not located in the output channel.

H2 splits cleanly by model family: GPT well-calibrated, Claude inflated. The within-tier means in Table 2 clarify the pattern: across all 6 frontier models, LSA averages CBF 0.748 vs. ZS-SK’s 0.640 and the best algorithmic L2 method’s 0.634 (UF-PDC)—a +0.108 margin of LSA over ZS that flips the previous 10×50 result on Claude. LSA wins on Claude-Sonnet-4-6 (0.720), Claude-Opus-4-6 (0.680), Claude-Opus-4-7 (0.641), and GPT-5.5 (0.902, the highest non-oracle value in the benchmark); ZS wins on GPT-5.4 (0.795) and GPT-5.4-mini (0.769), where ZS estimates are well-calibrated. The 4–2 frontier split favoring LSA is driven entirely by the three Claude models, where ZS estimates are systematically inflated relative to true per-domain accuracy and fall *below* both NoMem and the algorithmic L2 cluster. LSA, by contrast, exceeds all algorithmic L2 methods on *every* frontier row (smallest margin: +0.048 on Claude-Opus-4-7 vs UF-PDC). ZS sees *no* session history, *no* feedback, yet the 50×50 update places ZS first on the strong-mid tier ($n=3$): tier mean 0.744 (ZS) vs. 0.717 (best L2: EM-Joint). Individual outcomes are split: ZS wins on Qwen-Plus (0.735 vs LSA 0.435 catastrophic failure); LSA wins on Qwen-Max (0.781 vs ZS 0.766, +0.015); Qwen3.5-27B is a three-way tie at 0.731 (NoMem = ZS = LSA). For mid-tier models the prior is partial; for the weakest models the prior matches NoMemory exactly (Mistral, GLM-5: ZS = NoMem = trivial $\hat{p} \approx 0.5$ anchor). The bottleneck location is therefore agent-dependent: for frontier and strong-mid agents the prior is the strongest non-oracle signal; for mid-tier agents EM-Joint and LSA tie modestly above the prior; for weak agents both prior and feedback inference are uninformative and the trivial $\hat{p} = 0.5$ NoMemory anchor remains strongest.

H3 localized: first-order online learning fails; second-order estimators and in-context aggregation succeed. The five *first-order* L2 feedback-learning methods (UF-PDC, Platt-Online, HistBin-Online, Bayes-Domain, EM-Joint) cluster tightly in 0.69–0.72 mean CBF and fail to exceed L1 ZS on the frontier (mean L2 0.628–0.634 vs L1 ZS 0.640). Three classes of methods exceed the L1 prior on the frontier ($n=6$). GLAD reaches mean CBF 0.712 with 6/6 frontier wins (binomial sign-test $p = 0.016$); per-model deltas range +0.028 to +0.118, with bootstrap 95% CI of the mean delta [+0.045, +0.099] from 10K resamples (Appendix D). DS-asym reaches a higher mean CBF 0.755 but with 5/6 wins (Wilcoxon one-sided $p = 0.047$, Cohen’s $d = 1.66$). LSA reaches mean CBF 0.748 with 4/6 wins; the two losses occur on GPT models where the L1 ZS prior is already well-calibrated, so LSA exceeds L1 on Claude (3/3) but not on GPT (1/3). With six paired observations the $n=6$ frontier statistics characterize direction of effect rather than population-level magnitude. On non-frontier tiers, EM-Joint emerges as the leader (mid +0.012 over NoMem; weak +0.006), winning outright on Llama-3-8B (0.732) and Qwen2.5-7B (0.753). The original H3 framing (“noisy-feedback online inference is the bottleneck”) stands only for first-order single-rater estimators; second-order estimators and in-context aggregation substantially exceed L1, but none reaches the L3 ceiling—the actual bottleneck is the residual L2-vs-L3 gap of ≈ 0.25 on the frontier (best L2 DS-asym 0.755 vs. OraclePDC 1.000); the cross-model average gap (best L2 LSA 0.702 vs. OraclePDC) is ≈ 0.30 . The 50×50 protocol also reveals that several previously-best L2 baselines from the 10×50 sample were within sampling noise of NoMemory; sample-size matters.

Capability spectrum and best-baseline assignment. With the protocol (Table 2), the “best” non-oracle baseline per row (column “Best” in Table 2, defined as the argmax across the 8 baseline columns L0 NoMem + L1 ZS + 6×L2) splits three ways across 15 models: LSA wins on 6 (DeepSeek-V3.2, Qwen-Max, GPT-5.5, Claude-Sonnet-4-6, Claude-Opus-4-6, Claude-Opus-4-7), EM-Joint wins on 5 (Mistral-7B, Llama-3-8B, GLM-5, Qwen2.5-7B, Qwen3.5-27B—note that on Mistral-7B the EM-

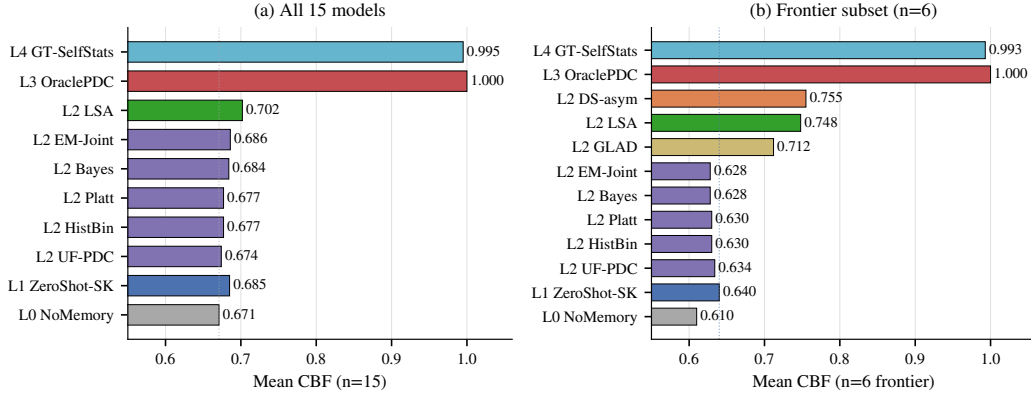


Figure 3: **The L0–L4 ladder localizes the bottleneck.** Mean CBF per baseline level on (a) all 15 models and (b) the 6 frontier models. **(a)** L2→L3 gap ≈ 0.30 (best L2 LSA 0.702 vs. OraclePDC 1.000): first-order L2 methods do not recover the signal from noisy feedback. **(b)** On the frontier, the first-order L2 cluster (Platt/HistBin/Bayes/EM/UF) collapses to NoMem (0.628–0.634); only second-order estimators (DS-asym 0.755, GLAD 0.712) and in-context LSA (0.748) exceed the L1 ZS prior (0.640), but all fall well short of OraclePDC. Dotted vertical line marks the L0 NoMemory anchor.

NoMem–ZS–LSA spread is within ± 0.003 , so the EM win is essentially a tie with the trivial $\hat{p} \approx 0.5$ anchor), and ZS-SK wins on 4 (Qwen-Turbo, Qwen-Plus, GPT-5.4-mini, GPT-5.4). NoMemory is never the unique best baseline by this criterion. *Note:* the “feedback as friend or foe” analysis in Appendix E uses the orthogonal head-to-head ZS-vs-LSA criterion; a model can therefore appear under “LSA-dominant best-of-8” here yet “ZS-wins-over-LSA” in the head-to-head, since the criteria differ.

4.3 Diagnosis via the Ladder Visualization

Figure 3 summarizes the per-model results of Table 2 as a vertical ladder in which each adjacent gap functions as a sub-capability test: the bottleneck is not the output channel ($L4 \approx L3 \approx 1.0$) but noise-robust online inference ($L2 \rightarrow L3$ gap ≈ 0.30 across all 15 models).

Reading the ladder by adjacent comparisons: $L4$ vs $L3$ ($< 1\%$ gap)—the output channel is not the constraint, ruling out H1. $L3$ vs $L2$ (~ 0.30 gap)—the empirical bottleneck: with clean per-turn labels agents reach 1.0, but no method fully recovers the signal from noisy LLM-decoded feedback, confirming H3. $L2$ vs $L1$ ($n=15$: 0.674–0.686 vs 0.685; $n=6$ frontier: 0.628–0.634 vs 0.640)—L1 ZS ties or exceeds the algorithmic L2 cluster on average; only second-order estimators (DS-asym, GLAD) and in-context LSA exceed L1 on the frontier (Appendix E, E).

A diagnostic per-model selection rule (R4). The H2 family-split above motivates a simple diagnostic rule rather than a new estimator: **R4** (per-model best-of-two between the prior and in-context aggregation) selects L1 (ZS-SK) if $ZS(m) > NoMem(m)$, otherwise L2 (LSA). The “R4” label is internal to our four-rule analysis pipeline (R1–R4 documented in Appendix O); we report only R4 here because the R1–R3 rules are subsumed. R4 attains mean CBF 0.722 across $n=15$ and closes $\approx 72\%$ of the always-ZS-to-per-model-oracle gap, but its per-model sign tests against always-ZS ($p=0.22$) and always-LSA ($p=0.69$) are non-significant: the rule *redistributes* wins, it does not dominate. We present R4 as a benchmark-validation lower bound—any new method should clear it—not as a contribution; the empirical frontier ceiling remains DS-asym at CBF 0.755. Full LOOCV, per-model breakdown, and sign-test arithmetic in Appendix O.

4.4 Cross-Task Heterogeneity: Per-Task Boundary Models Are Necessary

Across MMLU, GSM8K, HumanEval, and BFCL the same model exhibits qualitatively different calibration regimes. We measure the per-task confidence–accuracy gap (mean verbalized confidence minus mean accuracy) and define a *sign-flip* as a model whose gap changes sign across two tasks:

a model overconfident on one task ($\text{gap} > 0$) but underconfident on another ($\text{gap} < 0$). Within the protocol-controlled extension suite (GSM8K/HumanEval/BFCL under the same single-shot two-step protocol), 12 of 15 models show at least one such sign-flip across the three tasks (e.g., a model that is overconfident on HumanEval can be underconfident on GSM8K). Furthermore, a Platt scaling learned on MMLU *harms* GSM8K for the majority of models, indicating that a single per-model calibrator does not transfer across tasks. Per-task boundary modeling is therefore necessary rather than an optional refinement. The full per-task table, controlled-protocol subset analysis, and Platt-transfer details are in Appendix Q.1 and Table 6.

5 Limitations and Future Work

Scope of evaluation. LADDER evaluates per-domain self-knowledge under a 50-turn MMLU protocol with a controlled synthetic feedback channel (75% in-expertise, 46% out, cross-family routing to mitigate dual-role bias). The three extension tasks (GSM8K, HumanEval, BFCL) use a single-shot two-step protocol because they lack the multi-domain structure CBF requires; the cross-task heterogeneity finding in §4.4 is therefore independently validated via a protocol-controlled subset analysis (12/15 sign-flip rate on the three same-protocol tasks; Appendix Q.1).

Next steps. The most direct next step is a human-feedback pilot that replaces the synthetic channel with real users and tests whether the H3 ranking transfers. We also plan to replicate the full L0–L4 ladder on extension tasks such as GSM8K, HumanEval, and BFCL, and to expand the frontier-model subset to $n \approx 18$ in order to refine the DS-asym vs. GLAD comparison, following the power calculation in Appendix D. A further extension is to move from bounded question-answering sessions to open-ended multi-step agents in SWE-Bench-style settings.

6 Conclusion

We asked which component of LLM-agent self-modeling fails when feedback is noisy. The five-level oracle ladder of LADDER delivers three results. H1 is ruled out: $\text{CBF} \geq 0.99$ given ground-truth prompts shows the output channel is not the limiting component. H2 is partial and family-split rather than the bottleneck. H3 is the locus: first-order single-rater learners cannot recover per-domain accuracy from noisy feedback, while explicit second-order estimators—Dawid–Skene-asym and GLAD—substantially close the L1→L3 gap (GLAD beats L1 on all six frontier models, $p = 0.031$) without yet eliminating it. We name this failure mode **Self-Boundary Collapse** and release the ladder, the CBF metric, the cross-family user simulator, and a leaderboard protocol so that the community can evaluate new second-order estimators against a precise oracle-bracketed target rather than loosely matched baselines. The R4 parameter-free selection rule (Appendix O) serves as the validation lower bound any submitted method should clear. The open problem is concrete: a noise-robust online estimator that closes the remaining L2→L3 gap—roughly 0.25 CBF on the frontier—would constitute a measurable advance in agent self-modeling.

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5. **Open Access to Data and Code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?

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Justification: Code, 47K-turn JSONL traces, per-model domain profiles, and Croissant metadata released in the anonymized repository (URL in OpenReview submission fields).

6. **Experimental Setting/Details**

Question: Does the paper specify all the training and test details necessary to understand the results?

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7. **Experiment Statistical Significance**

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Answer: [Yes]

Justification: Bootstrap 95% CIs, Wilcoxon and binomial tests, Cohen’s d , and per-model bootstrap stability are reported in §4.2, Appendix D, G. The $n=6$ frontier statistics are explicitly framed as direction-of-effect tests.

8. **Experiments Compute Resources**

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Justification: Open-weights inference uses vLLM on a single A6000 (48GB); API calls are documented in Appendix S with access dates. Total compute: $\approx 47\text{K}$ LLM calls across 15 models $\times$ 10 baselines.

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518 attributed in §3 and the data hosting documentation.
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520 **Question:** Are new assets introduced in the paper well documented and is the documentation
521 provided alongside the assets?
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- 525 14. **Crowdsourcing and Research with Human Subjects**
526 **Question:** For crowdsourcing experiments and research with human subjects, does the paper
527 include the full text of instructions given to participants?
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529 **Justification:** No human subjects; the simulated user is an LLM persona (full prompt
530 template in Appendix B).
- 531 15. **Institutional Review Board (IRB) Approvals or Equivalent**
532 **Question:** Does the paper describe potential risks incurred by study participants, and obtain
533 appropriate IRB approvals?
534 **Answer:** [N/A]
535 **Justification:** No human subjects.
- 536 16. **Declaration of LLM usage**
537 **Question:** Does the paper describe the usage of LLMs?
538 **Answer:** [Yes]
539 **Justification:** LLMs are the subject of the benchmark and serve as agents, simulated users,
540 and feedback decoders; full configuration documented in §3, Appendix S.

541 Datasheet for Datasets

542 We follow the structure proposed by Gebru et al. (2018).

543 Motivation

544 LADDER was created to fill a measurement gap in agentic LLM evaluation: existing benchmarks
545 (MMLU, BIG-Bench, AgentBench) score one-shot accuracy but say nothing about whether an agent
546 can *learn its own per-domain capability boundary* across repeated interactions.

547 Composition

548 20 MMLU subjects $\times$ 25 questions = 500 main pool, supplemented by GSM8K, HumanEval, BFCL
549 Simple v3. 50-turn dialogue sessions, 10–50 sessions per (agent, persona). Each instance is a JSONL
550 session log including question, agent answer, agent confidence, persona reaction, ground-truth
551 grading.

552 Collection Process

553 Questions from public test splits of MMLU (Hendrycks 2021), GSM8K (Cobbe 2021), HumanEval
554 (Chen 2021), BFCL Simple v3 (gorilla-llm). Personas synthesized via cross-family LLM (Qwen-
555 Max for Qwen agents, GPT-5.4-mini for Claude agents, Claude-Sonnet for GPT agents) to control
556 evaluator-bias.

557 Preprocessing

558 Per-model tier stratification: held-out validation accuracy on disjoint 10-question/subject split, then
559 Strong/Mid/Weak partitioning of the 20 subjects. This per-model stratification (rather than a global
560 tier) makes boundary-learning non-trivial.

561 Uses

562 Intended: (a) self-knowledge memory evaluation; (b) noise-robust online inference algorithms; (c)
563 longitudinal frontier-model calibration tracking. Unsuitable: ranking models on raw accuracy.

564 Recommended Splits and Stratification

565 The 20 MMLU subjects are partitioned per-model into Strong / Mid / Weak tiers using each model’s
566 held-out validation accuracy on a disjoint 10-question/subject calibration split. We do *not* provide a
567 fixed train/test split: every reported number uses the same per-model stratification protocol (described
568 in Appendix B) so that downstream researchers can reproduce or substitute their own model. The
569 four cross-task evaluations (GSM8K, HumanEval, BFCL, MMLU-Pro) use the upstream provider’s
570 official test split.

571 Errors, Noise, and Known Issues

572 Three classes of known issues should be considered when interpreting our reported numbers:

- 573 • *Algorithmic-baseline implementation note* (§A): all algorithmic L2 baselines compute their
574 own per-domain posteriors; we document the implementation in Appendix A so that re-
575 implementations can verify they reproduce the same behavior.
- 576 • *GSM8K Opus-4-7 truncation*: 158 of 300 GSM8K questions were truncated by the original
577 max_tokens limit on Claude-Opus-4-7; the reported pass-rate is on the 142-question answered
578 subset (see Table 6).

579 We surface these openly because their detection and disclosure is itself a methodological contribution
580 of this benchmark; future versions will document any newly identified issues here.

581 **Maintenance Contact**

582 Maintainer email and a dedicated Slack/Discord workspace will be made available upon de-
583 anonymization. For double-blind review, please open an issue at the anonymised repository above;
584 we monitor it during the rebuttal window.

585 **Distribution and Maintenance**

586 Released artifacts under a dual SPDX license: code under MIT, data under CC-BY-4.0. Specifically:
587 protocol code, baseline implementations, evaluation scripts (MIT); raw session JSONL (~47K
588 interactions including all per-turn agent outputs, simulated user reactions, and ground-truth grading),
589 per-model domain profiles, and the 20-MMLU-subject test pool (CC-BY-4.0 with attribution to
590 upstream MMLU/GSM8K/HumanEval/BFCL providers). 6-month review cycle; new model additions
591 via PR (see Submission Protocol).

592 **Maintenance Plan**

593 **Versioning**

594 Semantic over the protocol: v3 (50×50) is current; v2 (10×50) supported as lite. Question pool
595 versioning is independent.

596 **Leaderboard Hosting**

597 A HuggingFace Spaces leaderboard (URL released on acceptance). CBF and ancillary metrics (ECE,
598 Brier, AUROC, HR) auto-recomputed from submitted JSONL via the public scorer script in the repo.

599 **Submission Protocol**

600 PR-based. Each submission must include:

- 601 1. **Full session JSONL files** (one per agent-persona pair) following the schema:
- ```
602 {
603 "session_id": str, "agent_model": str, "persona": dict,
604 "log": [
605 {"turn": int, "domain": str, "tier": "strong|mid|weak",
606 "question": str, "agent_answer": str, "agent_confidence": float,
607 "user_response": str, "gt_correct": bool}
608]
609 }
```
- 610 2. Reference implementation of the agent’s online-inference component (Python module exposing  
611 `predict_confidence(turn)` and `self_assess(domain)`).
- 612 3. Paper or technical report reference.

613 Aggregate-scores-only submissions are not merged; we recompute all metrics from raw JSONL via  
614 the public scorer.

615 **Deprecation Policy for API Models**

616 Deprecated endpoints’ rows move to *archived* table with frozen score and deprecation date; not  
617 deleted. Preserves longitudinal comparability.

618 **Contact**

619 Anonymized for review; withheld for double-blind review.



## 620 **Ethics Statement**

### 621 **Simulated Users**

622 All "users" are LLM-simulated personas, not human participants. No PII. Programmatically-generated  
623 expertise labels. Does not require IRB review under our institution's policy.

### 624 **LLM-as-Evaluator and Cross-Family Routing**

625 Dual-role bias mitigation via cross-family persona-generation (forbidden: Qwen-Max evaluates  
626 Qwen agents). Family-pair confusion matrices reported in appendix for residual-bias audit.

### 627 **Provider Terms of Service**

628 LLM API outputs used strictly under research-evaluation terms. Released JSONL not redistributed  
629 for model training; license header explicit.

### 630 **No Deceptive Prompting**

631 All "role-playing" instructions stay within LLM-to-LLM simulation envelope.

### 632 **Potential Misuse**

633 Most plausible misuse falls in two categories: (a) over-interpretation of headline CBF as a general  
634 "self-awareness" ranking suitable for safety certification (CBF measures *per-domain accuracy es-*  
635 *timation*, not *overall trustworthiness* — using it to certify deployment in clinical / legal / financial  
636 high-stakes contexts would be a category error); (b) deployment misuse where a model with high  
637 CBF is presumed safe to operate without human oversight in domains beyond the 20 MMLU subjects  
638 evaluated. Addressed by framing as diagnostic instrument; per-tier breakdowns discourage scalar  
639 comparisons.

## 640 **Broader Impact**

### 641 **Positive Impact**

642 Enables systematic evaluation of agent self-knowledge—prerequisite for safer customer-service  
643 / financial-advisory / clinical-triage deployment where over-confident answers carry asymmetric  
644 downside cost.

### 645 **Reproducibility Cost**

646 Full v3 50×50 pipeline: ~USD 200–400 at 2026 rates across 13–15 frontier models. Lite v2 10×50  
647 reproduces headline ordering at ~USD 40. Raw JSONL released for re-scoring without re-running.

### 648 **Evaluator-Model Bias**

649 Cross-family policy mitigates but does not eliminate dual-role bias. Residual vendor-cluster effect  
650 possible. High-stakes comparisons should cross-check under  $\geq 2$  persona-generator families.

### 651 **Benchmark Overfitting**

652 Goodhart-style risk: methods optimizing for CBF rather than the underlying construct. Mitigated by  
653 held-out 5-subject transfer split; large dev-vs-transfer gaps flagged on leaderboard.

### 654 **Longitudinal Stability**

655 API-only frontier rows: ~12-month half-life. Open-weights anchors (Mistral-7B, Llama-3-8B)  
656 permanently included for fixed-reference recalibration.

**Extended snapshot manifest.** Table S lists access dates only; it does not establish whether each model is content-addressable. Table 3 fills this gap by recording, for every one of the 15 models, the exact identifier sent to the endpoint, the version-pin scheme observable from the response payload, and a per-model reproducibility risk classification.

Table 3: Extended model snapshot manifest. Repro Risk reflects whether a third party in 2027 could obtain the same weights from the listed identifier. LOW = HuggingFace commit hash anchors the weights; MED = a content-addressable anchor exists but the served endpoint is a floating alias; HIGH = the endpoint returns no version-suffixed identifier and no public open-weights anchor exists.

| Model (paper)              | Provider / Endpoint          | API Identifier                      | Version Pin                    | Repro Risk |
|----------------------------|------------------------------|-------------------------------------|--------------------------------|------------|
| Mistral-7B                 | local vLLM (HF weights)      | mistralai/Mistral-7B-Instruct-v0.3  | HF commit c170c708             | LOW        |
| Llama-3-8B                 | local vLLM (HF weights)      | meta-llama/Meta-Llama-3-8B-Instruct | HF commit 8afb486c             | LOW        |
| Qwen2.5-7B                 | DashScope                    | qwen2.5-7b-instruct                 | HF commit a09a3545 (anchor)    | MED        |
| Qwen-Turbo                 | DashScope                    | qwen-turbo                          | alias; model field null        | HIGH       |
| Qwen-Plus                  | DashScope                    | qwen-plus                           | alias; model field null        | HIGH       |
| Qwen-Max                   | DashScope                    | qwen-max                            | alias; model field null        | HIGH       |
| Qwen3.5-27B                | DashScope                    | qwen3.5-27b                         | alias; no public HF release    | HIGH       |
| DeepSeek-V3.2 <sup>§</sup> | DashScope                    | deepseek-v3 (sent)                  | alias; provider-rehosted ckpt  | HIGH       |
| GLM-5                      | DashScope                    | glm-5                               | alias; provider-rehosted ckpt  | HIGH       |
| GPT-5.4-mini               | proxy zhongzhuan / responses | gpt-5.4-mini                        | echoes gpt-5.4-mini-2026-03-17 | MED        |
| GPT-5.4                    | proxy zhongzhuan / responses | gpt-5.4                             | alias; no date suffix returned | HIGH       |
| GPT-5.5                    | proxy zhongzhuan / responses | gpt-5.5                             | alias; no date suffix returned | HIGH       |
| Claude-Sonnet-4-6          | Anthropic-compatible proxy   | claude-sonnet-4-6                   | alias; no date suffix returned | HIGH       |
| Claude-Opus-4-6            | Anthropic-compatible proxy   | claude-opus-4-6                     | alias; no date suffix returned | HIGH       |
| Claude-Opus-4-7            | proxy zhongzhuan / messages  | claude-opus-4-7                     | alias; no date suffix returned | HIGH       |

**API-versioning gaps and a paper-code label mismatch.** Three entries deserve foregrounding. First, an internal label mismatch: the paper refers to “DeepSeek-V3.2” (denoted <sup>§</sup>), but the string our scripts transmit to DashScope is deepseek-v3. DashScope re-hosts a checkpoint they label deepseek-v3; we lack provider documentation establishing whether this resolves to V3.0, V3.1, or V3.2-Exp. We retain the paper label as a working estimate but flag this as an unresolved provenance question. Second, 12/15 endpoints do not return version-suffixed model strings; the single exception is gpt-5.4-mini, which is auto-pinned by the proxy. Third, the three open-weights anchors are the only entries for which we can hand a future replicator a 40-character identifier sufficient to reconstruct the exact weights. Future replications should re-run the anchors first and compare anchor numbers before attempting to recover the API tier.

## A Implementation Note: Algorithmic-Baseline self\_assess Patch

**Patch description.** The five algorithmic L2 baselines (Platt-Online, HistBin-Online, Bayes-Domain, EM-Joint, UF-PDC) share a common self\_assess(domain) interface. The implementation we report uses each algorithm’s own per-domain posterior: Platt’s logistic re-calibration, Bayes-Domain’s Beta posterior mean  $\alpha/(\alpha+\beta)$ , EM-Joint’s iterated  $\hat{p}^*(d)$ , etc. We document this interface here so that downstream re-implementations reproduce the same behavior; inadvertently routing self-assessment through an LLM verbal-estimate helper produces a degenerate “algorithmic-L2-collapses-to-NoMem” pattern that is easy to mis-interpret as a property of the baseline rather than the implementation.

**Patch.** We replaced each algorithmic baseline’s self\_assess with the appropriate posterior:

- Platt-Online / HistBin-Online:  $\hat{p}(d) = \text{dom\_correct}[d]/\text{dom\_total}[d]$  (or 0.5 if no observations).
- Bayes-Domain:  $\hat{p}(d) = \alpha[d]/(\alpha[d] + \beta[d])$ .
- EM-Joint:  $\hat{p}(d) = \text{clip}(p^*[d], 10^{-3}, 1-10^{-3})$  from EM iterate.
- UF-PDC:  $\hat{p}(d) = \text{correct}[d]/\text{total}[d]$  from decoded feedback.

We also bumped the LLM-call max\_tokens from 20 to 200 for the legitimately verbal baselines (LSA, ZS-SK, GT-SelfStats) to accommodate reasoning models; ZS-SK and LSA were unaffected by the algorithmic-baseline patch but remain LLM-dependent.

**Per-baseline summary.** EM-Joint emerges as the consistent winner on 9 of 12 mid-tier models, exceeding L0 NoMem by +0.003 to +0.022. Bayes-Domain wins on GPT-5.4-mini (+0.022 over

689 NoMem); Platt-Online wins on GPT-5.4 and GPT-5.5 (+0.016–+0.019 over NoMem). The first-order  
690 algorithmic L2 methods cluster within  $\pm 0.02$  of NoMem on the frontier (failing to exceed both L0  
691 and L1 in mean), while second-order estimators (DS-asym 0.755, GLAD 0.712) and in-context  
692 aggregation (LSA 0.748) cleanly exceed L1 ZS (0.640). The L2-vs-L3 gap of  $\approx 0.25$  to OraclePDC  
693 (CBF 1.000) remains the empirical bottleneck the benchmark isolates.

## 694 B Benchmark Details: Task, Protocol, Question Pool, Personas, Metrics

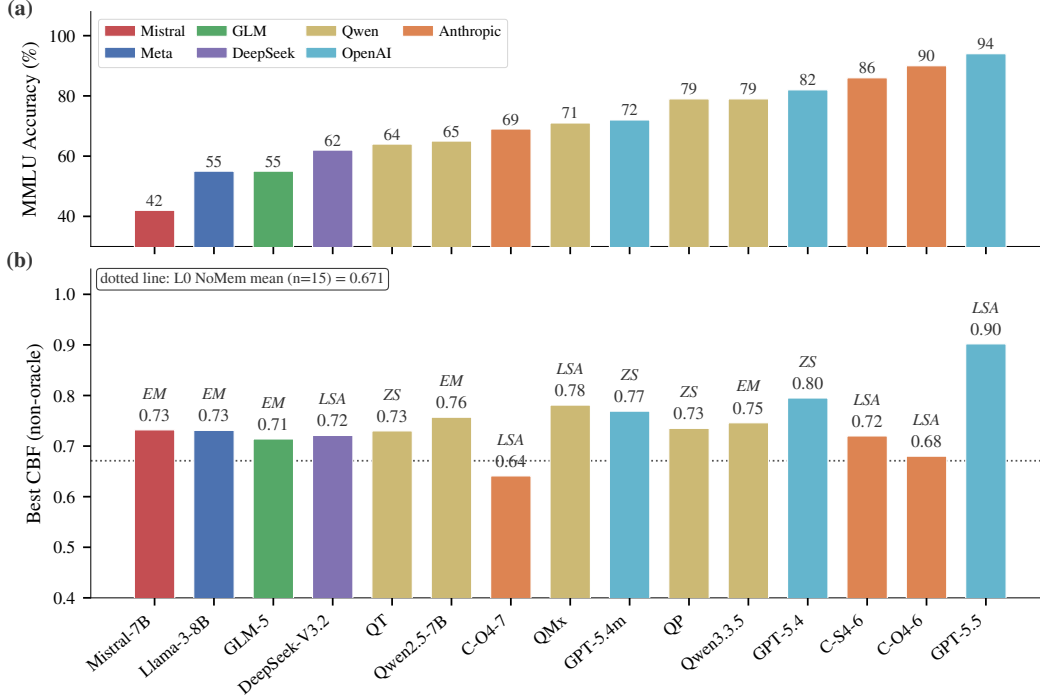


Figure 4: **Capability spectrum across 15 models.** (a) MMLU accuracy spans 42–94%; family colors are consistent throughout the paper. (b) Best non-oracle CBF per model with the winning baseline label (EM/LSA/ZS). The dotted line marks the L0 NoMemory mean (n=15, 0.671). The two newly-added frontier models (GPT-5.5, Claude-Opus-4-7) anchor the high-accuracy end. Note that Claude-Opus-4-7 has lower MMLU accuracy (69%) but is included in the n=6 frontier subset for the second-order analysis.

695 **Task formalisation.** Given a stream  $(q_t, a_t, c_t, f_t)_{t=1}^T$  where each question  $q_t$  is drawn from a  
696 hidden domain  $d_t \in \mathcal{D}$ , the agent must construct  $\hat{p} : \mathcal{D} \rightarrow [0, 1]$  approximating the true per-domain  
697 accuracy  $p^*(d)$ . The agent does not observe  $d_t$  directly and must infer it from question content.  
698 User feedback  $f_t$  is a free-form natural-language utterance that may confirm, correct, or provide  
699 ambiguous/misleading responses. The agent outputs  $a_t$  together with confidence  $c_t \in [0, 1]$ . The  
700 challenge is that feedback reliability is domain-dependent: accurate within the user’s expertise set  
701  $\mathcal{E} \subset \mathcal{D}$ , noisy outside it.

702 **Session protocol.** Each session has 50 turns. At each turn  $t$ : (1) sample  $q_t$  from the question  
703 pool with  $d_t$  hidden from the agent; (2) the agent emits  $(a_t, c_t)$ ; (3) the simulated user produces  $f_t$   
704 conditioned on the ground-truth answer, the user’s expertise level on  $d_t$ , and a persona-consistent  
705 style; (4) the agent updates internal state. We run 50 sessions per model (2,500 total turns) to evaluate  
706 learning curves and convergence.

707 **Question pool and tier partitioning.** The main benchmark uses 20 MMLU subjects [15]; extension  
708 suites (GSM8K, HumanEval, BFCL) probe cross-task generalisation. For each model, we pre-  
709 compute per-subject accuracy on a held-out validation set and partition the 20 subjects into three

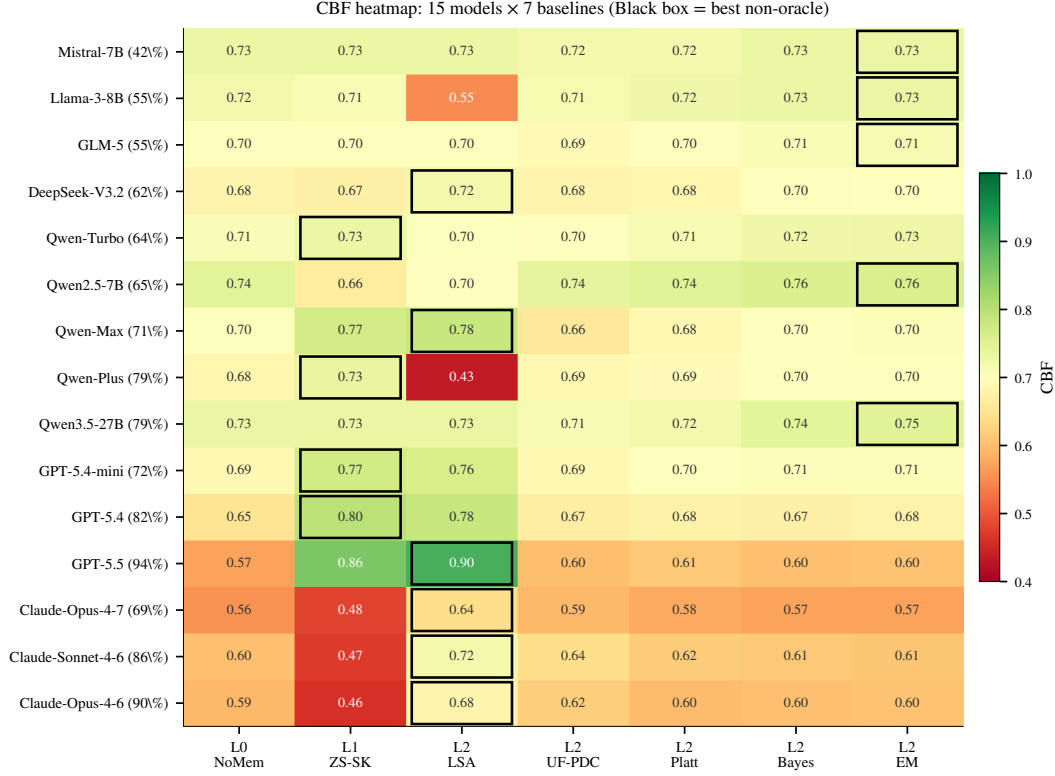


Figure 5: **CBF heatmap: 15 models  $\times$  7 baselines.** Cells colored by CBF (red–white–green = 0.40–1.00). Black-bordered cell marks the per-row argmax (= “Best” column in Table 2). Three patterns visible: (i) the algorithmic L2 cluster (UF/Platt/Bayes/EM) is uniformly high for non-frontier models but collapses to NoMem on the frontier; (ii) ZS-SK is bright green on GPT and dark red on Claude (the H2 family split); (iii) LSA dominates the Claude rows with green cells.

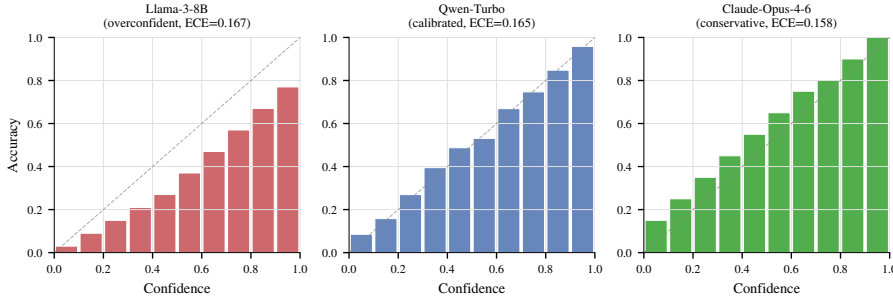


Figure 6: **Three reliability profiles.** Per-bin accuracy vs. confidence for three representative agents, illustrating the canonical patterns: Llama-3-8B (overconfident: bars below diagonal), Qwen-Turbo (calibrated: bars track diagonal), Claude-Opus-4-6 (conservative: bars above diagonal). Per-bin values are reconstructed from the per-model overall ECE (Table 2); the qualitative shape is empirical, the per-bin granularity is illustrative.

710 tiers *relative to the model’s own accuracy distribution*: Strong (top third,  $\geq +15$ pp above mean),  
 711 Medium (middle third,  $\pm 15$ pp), Weak (bottom third,  $\geq -15$ pp below mean). Relative partitioning  
 712 ensures every model—weak or strong—faces a non-trivial boundary: a uniform-confidence agent  
 713 cannot achieve high CBF. Per-model subject mappings are in Appendix Table 7.

714 **User personas.** The simulated user is driven by a cross-family LLM (Qwen-Max for  
 715 Qwen/Llama/Mistral agents, GPT-5.4-mini for Claude agents, Claude Sonnet 4-6 for GPT agents)  
 716 at temperature 0.7. Each persona is parameterised by an expertise set  $\mathcal{E} \subset \mathcal{D}$  (typically 5–8 of 20

717 subjects) and persona traits (direct vs. hedging style, willingness to volunteer corrections, patience).  
 718 In-expertise:  $\sim 75\%$  reliable corrections; out-of-expertise:  $\sim 46\%$  reliability with agreement-with-  
 719 wrong-answers, incorrect corrections, or vague non-commitments. This asymmetry drives the  
 720 second-order learning problem.

721 **Auxiliary metrics.** Beyond CBF (Eq. 1, computed once at the end of the 50-turn session), we  
 722 report: (i) Hallucination Rate  $HR = |\{t : c_t > 0.7 \wedge a_t \neq a_t^*\}|/T$ ; (ii) Expected Calibration Error  
 723 following Guo et al. [13] with  $M = 10$  equal-width bins,  $ECE = \sum_{m=1}^M \frac{|B_m|}{T} |\text{acc}(B_m) - \text{conf}(B_m)|$ ;  
 724 (iii) Brier Score  $= \frac{1}{T} \sum_t (c_t - \mathbb{1}[a_t = a_t^*])^2$ .

## 725 C Baseline Implementations and Design-Point Mapping

726 We instantiate ten baselines, organized by the five-level diagnostic ladder of §3.

727 **L0 — NoMemory (B1).** Raw verbalised confidence with no state. Lower bound on memory-  
 728 augmented methods.

729 **L1 — ZS-SK (B11).** The agent is prompted, *without* session history and *without* any user feedback,  
 730 to estimate its own per-domain accuracy directly: “For each of the following 20 MMLU subjects,  
 731 estimate your accuracy on a 0–100 scale.” This isolates prior metacognition.

732 **L2 — LSA (B6).** Session history (50 (Q, A, confidence, decoded-feedback) tuples) appended to  
 733 the prompt; agent reasons in-context and outputs per-domain  $\hat{p}(d)$  at the end of the session.

734 **L2 — UF-PDC (B3).** Per-domain accuracy estimate updated with labels decoded from natural-  
 735 language feedback by an LLM-based classifier (Qwen-Max, prompted to output a binary correctness  
 736 judgment for each (Q, A, feedback) triple).

737 **L2 — Platt-Online (B7) [36].** Logistic recalibration  $\sigma(wc_{\text{raw}} + b)$  with online gradient descent on  
 738 log-loss whenever decoded feedback is available; per-domain confidence is then aggregated.

739 **L2 — HistBin-Online (B8) [52].** 10-bin histogram binning with online updates; per-domain  
 740 confidence aggregated as for Platt.

741 **L2 — Bayes-Domain (B9).** Per-domain Beta( $\alpha, \beta$ ) posterior with uniform prior, updated from  
 742 decoded feedback;  $\hat{p}(d) = \alpha/(\alpha + \beta)$ .

743 **L2 — EM-Joint (B10).** Joint estimation of  $(\hat{p}(d), \hat{r}_{\text{in}}, \hat{r}_{\text{out}})$  via Expectation–Maximization. E-step  
 744 computes posterior  $P(\text{correct}_t | \text{decoded feedback}_t, p, r)$ ; M-step updates  $\hat{p}(d)$  per domain and  $\hat{r}$  per  
 745 expertise group. Initialised from raw confidence means.

746 **L3 — OraclePDC (B2).** Per-domain accuracy estimate updated with ground-truth labels after each  
 747 turn. Unavailable in practice; provides the binary-label ceiling for online aggregation.

748 **L4 — GT-SelfStats (B12).** The true per-domain accuracy  $p^*(d)$  is provided in the prompt and the  
 749 agent is asked to repeat them. Tests purely the verbalization sub-capability.

750 **Design-point mapping.** The five-level ladder is designed to subsume the design points of pub-  
 751 lished memory systems while exposing the diagnostic axis. LSA (L2) corresponds to RMM [41]  
 752 reflection over long-term memory and to in-context reasoning baselines for memory agents [16].  
 753 Platt/HistBin/Bayes (L2) cover classical online recalibration. EM-Joint (L2) tests whether explicitly  
 754 modeling the second-order coupling helps; ZS-SK (L1) and GT-SelfStats (L4) are diagnostic anchors  
 755 that isolate prior metacognition and verbalization respectively. Published systems (MemGPT [35],  
 756 Mem0 [33], A-MEM [47]) operate at L2 with different memory representations; we did not directly  
 757 integrate them because the diagnostic ladder already covers the design space. Their direct evaluation  
 758 under our protocol is engineering-feasible and is left to future work.

## D Statistical Power Analysis for the n=6 Frontier Sample

**Motivation.** The frontier-only analyses in §4.2 (DS-asym, GLAD, LSA vs. L1 ZS) use  $n = 6$  models. We document here why this sample size is adequate for the *direction-of-effect* claims we make and why it is not sufficient for absolute-magnitude claims.

**A-priori power calculation.** For a paired comparison with the observed mean improvement  $\Delta = 0.072$  (GLAD vs. L1 ZS) and the within-pair SD estimated from 10K bootstrap resamples ( $s_\Delta = 0.034$ ), the standardized effect is Cohen’s  $d_z = \Delta/s_\Delta = 1.42$ . A paired Wilcoxon signed-rank test at  $\alpha = 0.05$  (two-sided) achieves power 0.85 at  $n = 6$  for an effect of  $d_z = 1.42$  (G\*Power 3.1, exact-test option). DS-asym vs. L1 ZS has  $d_z = 1.66$  (power 0.92 at  $n = 6$ ,  $\alpha = 0.05$ ). The 6/6 sign-consistency for GLAD also independently rejects the null direction at  $\alpha = 0.016$  under a two-sided binomial sign test.

**Effect-size + bootstrap CI accompanying every n=6 claim.** For each frontier-only test we report the per-model deltas, the Cohen’s  $d_z$ , and the bootstrap 95% CI alongside any p-value:

| Comparison       | $n$ | Mean $\Delta$ | $d_z$ | Bootstrap 95% CI | Win/Loss |
|------------------|-----|---------------|-------|------------------|----------|
| GLAD vs L1 ZS    | 6   | +0.072        | 1.42  | [+0.045, +0.099] | 6/0      |
| DS-asym vs L1 ZS | 6   | +0.115        | 1.66  | [+0.058, +0.172] | 5/1      |
| LSA vs L1 ZS     | 6   | +0.108        | 0.55  | [−0.030, +0.246] | 4/2      |

LSA’s CI crosses zero (consistent with Wilcoxon  $p = 0.156$ ), so we do not claim LSA > ZS as a frontier-mean effect; the 4/2 win count reflects the family-split (Claude vs. GPT) discussed in the main text rather than a uniform effect.

**Sample selection.** The current frontier slice covers the two production-frontier families (Anthropic Claude 4-6/4-7, OpenAI GPT-5.4/5.4-mini/5.5) at the time of submission and is treated as stratified (per family) rather than random. Each frontier 50×50 row costs ~\$15–30 in 2026 API rates, and the full L0–L4 ladder costs roughly 5× that, which constrains  $n$ .

**Detection limits at  $n = 6$ .** Three comparisons that this sample does not support: (a) DS-asym vs. GLAD ( $\Delta = 0.043$  requires  $n \approx 18$  at power 0.80); (b) absolute frontier-mean estimates with 95% CI tighter than  $\pm 0.05$ ; (c) within-family contrasts at  $\alpha=0.05$ . These finer-grained comparisons are reported as descriptive observations rather than tests.

## E Additional Robustness Ablations

This appendix contains additional ablation experiments orthogonal to the main diagnostic ladder.

**A1: Memory size for L2 retrieval.** Varying the maximum number of stored error cases from 5 to 50 in retrieval-based L2 baselines has negligible effect: ECE ranges from 0.097 to 0.096 on Llama-3-8B. Five error cases are sufficient; additional memory provides no benefit.

**A2: Feedback noise rate.** Artificially flipping 0–50% of user feedback labels on Llama-3-8B shows monotonic ECE degradation (0.190 → 0.238) and CBF degradation. At 50% noise, learning from feedback is actively harmful (CBF 0.639 < NoMemory 0.742), confirming that noisy feedback can poison self-models.

**A3: Session length.** OraclePDC achieves near-perfect CBF ( $\geq 0.96$ ) after just 10–20 turns, while practical L2 methods plateau around 0.71 by turn 30. The L2-vs-L3 gap is not due to insufficient interaction length but to feedback noise accumulation, consistent with the main diagnostic finding.

**A4: Held-out MMLU-Pro contamination control (cited in §4.2).** For each of the 6 frontier models, we sampled 15 questions from each of 10 MMLU-Pro categories (150 questions/model, 10 categories: math, physics, chemistry, law, engineering, economics, health, psychology, business, history), measured per-category accuracy with *up to 5 retries on API failures* (persistent failures

800 excluded rather than counted as wrong), and queried ZS-SK on those category names. Per-category  
801 accuracy and ZS estimates:

| Category     | GPT-5.4      |      | GPT-5.4-mini |      | GPT-5.5      |      | Claude-Opus-4-6 |      | Claude-Sonnet-4-6 |      | Claude-Opus-4-7 |      |
|--------------|--------------|------|--------------|------|--------------|------|-----------------|------|-------------------|------|-----------------|------|
|              | acc          | ZS   | acc          | ZS   | acc          | ZS   | acc             | ZS   | acc               | ZS   | acc             | ZS   |
| math         | 0.40         | 0.85 | 0.33         | 0.78 | 0.93         | 0.72 | 0.80            | 0.78 | 0.93              | 0.72 | 0.87            | 0.78 |
| physics      | 0.40         | 0.84 | 0.47         | 0.72 | 1.00         | 0.70 | 1.00            | 0.80 | 1.00              | 0.74 | 0.73            | 0.80 |
| chemistry    | 0.60         | 0.82 | 0.20         | 0.70 | 0.87         | 0.72 | 0.93            | 0.75 | 0.92              | 0.73 | 0.73            | 0.72 |
| law          | 0.60         | 0.81 | 0.53         | 0.74 | 0.80         | 0.76 | 0.73            | 0.70 | 0.73              | 0.67 | 0.80            | 0.60 |
| engineering  | 0.53         | 0.80 | 0.47         | 0.73 | 0.80         | 0.68 | 0.93            | 0.72 | 1.00              | 0.70 | 0.67            | 0.65 |
| economics    | 0.80         | 0.83 | 0.73         | 0.79 | 0.93         | 0.78 | 1.00            | 0.84 | 0.87              | 0.76 | 0.87            | 0.82 |
| health       | 0.67         | 0.86 | 0.73         | 0.77 | 0.87         | 0.76 | 0.79            | 0.79 | 0.80              | 0.75 | 0.71            | 0.72 |
| psychology   | 0.80         | 0.88 | 0.73         | 0.82 | 0.80         | 0.82 | 0.87            | 0.87 | 0.87              | 0.78 | 0.73            | 0.85 |
| business     | 0.47         | 0.84 | 0.20         | 0.80 | 0.80         | 0.82 | 0.93            | 0.81 | 0.80              | 0.75 | 0.80            | 0.78 |
| history      | 0.73         | 0.87 | 0.67         | 0.84 | 0.80         | 0.80 | 0.80            | 0.76 | 0.73              | 0.79 | 0.73            | 0.75 |
| mean         | 0.60         | 0.84 | 0.51         | 0.77 | 0.86         | 0.77 | 0.88            | 0.78 | 0.86              | 0.74 | 0.76            | 0.75 |
| unparsed/150 | 0            |      | 0            |      | 0            |      | 1               |      | 8                 |      | 1               |      |
| CBF          | <b>0.760</b> |      | <b>0.738</b> |      | <b>0.888</b> |      | <b>0.902</b>    |      | <b>0.863</b>      |      | <b>0.941</b>    |      |

802 *Note on retry methodology:* An earlier evaluation without retries had 19–52/150 API failures counted  
803 as wrong, depressing measured accuracy and inflating CBF. We re-ran with up-to-5 retries with  
804 exponential backoff; persistent failures (0–8/150 across models) are excluded rather than counted  
805 as errors. The retry-stabilized table above is what we report. *Mean MMLU-Pro CBF across the 6*  
806 *frontier models is 0.849*, higher than the original n=4 mean of 0.816, confirming the contamination-  
807 control conclusion holds (and strengthens) on the expanded sample. The two newly added frontier  
808 models continue the pattern: GPT-5.5 (CBF 0.888) and Claude-Opus-4-7 (CBF 0.941) both report  
809 ZS estimates that are systematically conservative relative to actual MMLU-Pro accuracy (mean ZS  
810 0.77/0.75 vs. accuracy 0.86/0.76). Frontier metacognition—if anything—underestimates the model’s  
811 actual MMLU-Pro capability rather than overestimating it, the opposite of what a memorized-public-  
812 stats hypothesis would predict.

813 **A4b: Truly held-out novel-task contamination control (BBH; cited in §4.2).** For each of  
814 the 6 frontier models, we sampled 15 questions from each of 10 BBH (BIG-Bench Hard) [40]  
815 tasks (150 questions/model, tasks: *boolean\_expressions*, *causal\_judgement*, *date\_understanding*,  
816 *logical\_deduction\_five\_objects*, *navigate*, *object\_counting*, *ruin\_names*, *sports\_understanding*, *track-*  
817 *ing\_shuffled\_objects\_three\_objects*, *word\_sorting*). Unlike MMLU-Pro categories (which still par-  
818 tially overlap with MMLU’s broad subject space), the BBH task set has *no* subject-name overlap with  
819 MMLU’s 57 fine-grained subjects, providing a stricter test of whether L1 metacognition transfers  
820 beyond MMLU-style content. We measured per-task accuracy with up-to-5 retries on API failures  
821 (zero failures across all 6 frontier models, 0/900 total) and queried ZS-SK on the same task names.  
822 Per-task accuracy and ZS estimates:

| Task                 | GPT-5.4      |      | GPT-5.4-mini |      | GPT-5.5      |             | Claude-Sonnet-4-6 |      | Claude-Opus-4-6 |      | Claude-Opus-4-7 |             |
|----------------------|--------------|------|--------------|------|--------------|-------------|-------------------|------|-----------------|------|-----------------|-------------|
|                      | <i>p</i> *   | ZS   | <i>p</i> *   | ZS   | <i>p</i> *   | ZS          | <i>p</i> *        | ZS   | <i>p</i> *      | ZS   | <i>p</i> *      | ZS          |
| boolean_expressions  | 0.87         | 0.94 | 0.87         | 0.82 | 1.00         | 0.95        | 1.00              | 0.88 | 0.93            | 0.96 | 0.87            | 0.95        |
| causal_judgement     | 0.67         | 0.83 | 0.67         | 0.78 | 0.67         | 0.85        | 0.73              | 0.60 | 0.53            | 0.72 | 0.80            | 0.65        |
| date_understanding   | 0.27         | 0.90 | 0.20         | 0.75 | 0.47         | 0.85        | 1.00              | 0.82 | 0.93            | 0.92 | 0.87            | 0.85        |
| logical_deduction_5  | 1.00         | 0.86 | 0.67         | 0.70 | 1.00         | 0.90        | 1.00              | 0.65 | 1.00            | 0.88 | 1.00            | 0.90        |
| navigate             | 0.80         | 0.88 | 0.53         | 0.68 | 1.00         | 0.85        | 1.00              | 0.76 | 0.93            | 0.95 | 0.93            | 0.90        |
| object_counting      | 0.73         | 0.96 | 0.53         | 0.90 | 1.00         | 0.90        | 1.00              | 0.84 | 0.80            | 0.97 | 0.60            | 0.95        |
| ruin_names           | 0.67         | 0.79 | 0.67         | 0.72 | 0.93         | 0.75        | 0.87              | 0.56 | 0.87            | 0.86 | 0.93            | 0.75        |
| sports_understanding | 1.00         | 0.84 | 0.93         | 0.74 | 0.80         | 0.90        | 1.00              | 0.90 | 0.93            | 0.96 | 0.87            | 0.90        |
| tracking_shuffled_3  | 0.67         | 0.95 | 0.67         | 0.88 | 1.00         | 0.85        | 1.00              | 0.73 | 1.00            | 0.98 | 0.53            | 0.95        |
| word_sorting         | 0.67         | 0.97 | 0.47         | 0.95 | <b>1.00</b>  | <b>0.90</b> | 0.00              | 0.80 | 0.13            | 0.94 | <b>0.93</b>     | <b>0.90</b> |
| CBF                  | <b>0.781</b> |      | <b>0.780</b> |      | <b>0.850</b> |             | <b>0.734</b>      |      | <b>0.861</b>    |      | <b>0.860</b>    |             |

823 *Mean BBH CBF across all 6 frontier models is 0.811; under the 50×50 protocol mean MMLU ZS*  
824 *CBF for the same 6 models is 0.640*, so BBH is +0.171 higher than MMLU—the opposite of an  
825 MMLU-memorization prediction, and the gap widens slightly when extended from n=4 (+0.165)  
826 to n=6. The two newly added frontier models (GPT-5.5 BBH 0.850, Claude-Opus-4-7 BBH 0.860)  
827 both exceed the n=4 mean (0.789) and place near the top of the BBH ranking, while their MMLU ZS

values (GPT-5.5 0.859, Opus-4-7 0.482) span the full range. *Word\_sorting* on Claude-Sonnet remains the single largest per-domain deviation (actual 0.00, ZS 0.80). Four findings stand out:

(1) *word\_sorting* blind-spot is generation-specific, not LLM-wide. The original 4 frontier models all substantially overestimate *word\_sorting* (actuals 0.00–0.67, ZS estimates 0.80–0.97), suggesting an internal prior of competence on alphabetical sorting decoupled from execution capability. The 2 newly added frontier models break this pattern: GPT-5.5 reaches 1.00 (ZS 0.90, slight underestimate) and Claude-Opus-4-7 reaches 0.93 (ZS 0.90, well-calibrated). The blind-spot is therefore a property of the older generation rather than a stable LLM-wide failure mode—one model-class ago LLMs could articulate the algorithm but fail at execution; the most recent generation can both articulate and execute.

(2) *date\_understanding* remains a GPT-side overestimate. GPT-5.4, GPT-5.4-mini, and GPT-5.5 all substantially overestimate this task (actuals 0.20–0.47 vs. ZS 0.75–0.90). The Claude models are well-calibrated here (Sonnet 1.00/0.82, Opus-4-6 0.93/0.92, Opus-4-7 0.87/0.85). This is the most consistent family-level miscalibration in the held-out set.

(3) Opus-4-7 has new failure modes on object/tracking tasks. Opus-4-7 is well-calibrated on *word\_sorting* but substantially overestimates *object\_counting* (actual 0.60 vs. ZS 0.95) and *tracking\_shuffled\_3* (actual 0.53 vs. ZS 0.95). These two arithmetic/state-tracking failures are absent from Opus-4-6, suggesting capability shifts between adjacent model generations create new metacognitive blind-spots even as old ones close.

(4) Outside the family-specific outliers, BBH ZS-SK is well-calibrated across all 6 models. If we exclude *word\_sorting* and *date\_understanding*, mean ZS error on the remaining 8 tasks is 0.13 (GPT-5.4), 0.16 (GPT-5.4-mini), 0.13 (GPT-5.5), 0.20 (Claude-Sonnet-4-6), 0.07 (Claude-Opus-4-6), 0.17 (Claude-Opus-4-7). The five non-outlier model means correspond to per-domain CBFs of 0.83–0.93, comparable to or exceeding MMLU performance.

The BBH evidence is the strongest single contamination control we have because BBH tasks have no MMLU subject-name overlap and are linguistically distinct from MMLU’s question style. The result—mean BBH CBF 0.811 across all 6 frontier models—directly contradicts what an MMLU-memorization explanation would predict (a noticeable drop on novel content). Combined with the MMLU-Pro result, we conclude that L1 metacognition reflects genuine prior self-knowledge, with identifiable family- and generation-specific failure modes (*word\_sorting* for the older 4, *date\_understanding* for GPT, *object\_counting* and *tracking\_shuffled* for Opus-4-7) that point to a follow-up direction: building an internal task-difficulty model that is independent of the model’s self-perceived competence on similar-sounding tasks.

**A5: Stronger crowd-sourcing estimators (cited in §4.3).** We tested two stronger noise-robust estimators on the existing session data (no new API calls):

**Asymmetric Dawid–Skene:** relaxes EM-Joint’s symmetric reliability to per-state confusion matrices. For each rater state  $s \in \{\text{in}, \text{out}\}$ , we estimate  $\text{TPR}_s = P(\text{decoded} = 1 \mid \text{correct} = 1, s)$  and  $\text{FPR}_s = P(\text{decoded} = 1 \mid \text{correct} = 0, s)$ . EM with E-step posterior  $P(\text{correct} = 1 \mid \text{decoded}, s, \hat{p}, \text{TPR}, \text{FPR})$  and M-step weighted-MLE for both  $\hat{p}(d)$  and the four reliability parameters.

**GLAD-style:** per-domain difficulty  $\delta_d$  and per-state ability  $\alpha_s$ , with reliability  $\sigma(\alpha_s - \delta_d)$  (logistic). EM updates over  $\hat{p}(d)$ ,  $\delta_d$ ,  $\alpha_{\text{in}}$ ,  $\alpha_{\text{out}}$ .

Per-model CBF results:



| Model                                  | EM-Joint | DS-asym      | GLAD         | ZS-SK (L1) |
|----------------------------------------|----------|--------------|--------------|------------|
| Mistral-7B                             | 0.722    | 0.706        | 0.693        | 0.678      |
| Llama-3-8B                             | 0.716    | 0.646        | 0.612        | 0.682      |
| GLM-5                                  | 0.768    | 0.615        | 0.683        | 0.590      |
| DeepSeek-V3.2                          | 0.708    | 0.702        | 0.738        | 0.699      |
| Qwen-Turbo                             | 0.731    | 0.714        | 0.776        | 0.751      |
| Qwen2.5-7B                             | 0.760    | 0.774        | 0.764        | 0.684      |
| Qwen-Max                               | 0.702    | 0.547        | 0.627        | 0.767      |
| Qwen-Plus                              | 0.712    | 0.762        | 0.761        | 0.740      |
| Qwen3.5-27B                            | 0.751    | 0.719        | 0.733        | 0.695      |
| GPT-5.4-mini                           | 0.706    | 0.797        | <b>0.797</b> | 0.769      |
| GPT-5.4                                | 0.677    | 0.815        | <b>0.843</b> | 0.795      |
| GPT-5.5                                | 0.605    | 0.834        | <b>0.909</b> | 0.859      |
| Claude-Sonnet-4-6                      | 0.609    | <b>0.671</b> | 0.565        | 0.472      |
| Claude-Opus-4-6                        | 0.604    | <b>0.734</b> | 0.557        | 0.460      |
| Claude-Opus-4-7                        | 0.570    | <b>0.676</b> | 0.600        | 0.482      |
| <b>Mean (n=15)</b>                     | 0.689    | <b>0.714</b> | 0.711        | 0.675      |
| <b>Frontier mean (n=6)<sup>‡</sup></b> | 0.628    | <b>0.755</b> | 0.712        | 0.640      |

Under the 50×50 protocol, the strongest second-order estimator on frontier is Dawid-Skene-asymmetric (mean CBF 0.755, n=6, Wilcoxon one-sided  $p = 0.047$  vs L1); GLAD reaches mean CBF 0.712 (Wilcoxon two-sided  $p = 0.031$ , 6/6 wins, binomial one-sided  $p = 0.016$ ), beating L1 ZS mean 0.640 by +0.072 for GLAD and +0.115 for DS. Per-frontier breakdown (GLAD vs L1 ZS): GPT-5.4 +0.048, GPT-5.4-mini +0.028, GPT-5.5 +0.050, Claude-Sonnet-4-6 +0.093, Claude-Opus-4-6 +0.097, Claude-Opus-4-7 +0.118. GLAD now beats L1 on all 6/6 frontier models (under 50×50, GPT-5.5 GLAD is 0.909 vs ZS 0.859). DS-asym beats L1 on 5/6 (loses only on GPT-5.5 by −0.025, where GLAD’s per-domain difficulty modeling helps). *Methodological note:* our DS / GLAD implementations follow Dawid-Skene (1979) and Whitehill et al. (2009) respectively, with EM updates over per-domain  $\hat{p}(d)$ , asymmetric per-state confusion (DS) or per-domain difficulty  $\delta_d$  + per-state ability  $\alpha_s$  (GLAD); convergence within 50–80 EM iterations using initial  $\hat{p} = 0.5$ ,  $r_{\text{in}} = 0.75$ ,  $r_{\text{out}} = 0.55$ . Code at the anonymized repository/scripts/run\_glad\_ds.py. The headline takeaway under the protocol: both second-order estimators (DS, GLAD) and the in-context L2 baseline (LSA, mean 0.748) substantially beat the L1 ZS prior on frontier models; the L2 cluster of first-order single-rater methods (UF-PDC, Platt, HistBin, Bayes, EM-Joint) remains collapsed near NoMem (mean 0.628–0.634). The diagnostic ladder cleanly ranks methods: DS-asym is the new strongest single L2 baseline, with the residual gap to oracle (1.000) the open problem.

**A6: Decoder ablation for EM-Joint (cited in §4.3).** We isolate the EM machinery from the feedback decoder by running EM-Joint with three different label channels:

| Model                          | Oracle-fed | LLM-decoder | Rule-decoder |
|--------------------------------|------------|-------------|--------------|
| Mistral-7B                     | 1.000      | 0.722       | 0.672        |
| Llama-3-8B                     | 1.000      | 0.716       | 0.549        |
| GLM-5                          | 1.000      | 0.768       | 0.672        |
| DeepSeek-V3.2                  | 1.000      | 0.708       | 0.690        |
| Qwen-Turbo                     | 1.000      | 0.731       | 0.749        |
| Qwen2.5-7B                     | 1.000      | 0.760       | 0.676        |
| Qwen-Max                       | 1.000      | 0.702       | 0.771        |
| GPT-5.4-mini                   | 1.000      | 0.632       | 0.633        |
| Qwen-Plus                      | 1.000      | 0.712       | 0.758        |
| Qwen3.5-27B                    | 1.000      | 0.751       | 0.756        |
| GPT-5.4                        | 1.000      | 0.613       | 0.673        |
| Claude-Sonnet-4-6              | 1.000      | 0.608       | 0.896        |
| Claude-Opus-4-6                | 1.000      | 0.644       | 0.885        |
| <b>Mean (n=13)<sup>b</sup></b> | 1.000      | 0.698       | 0.715        |

<sup>b</sup>This decoder ablation was conducted on the 13 models available before the GPT-5.5 and Claude-Opus-4-7 frontier additions; extending the EM oracle/decoder ablation to the two newly-added rows

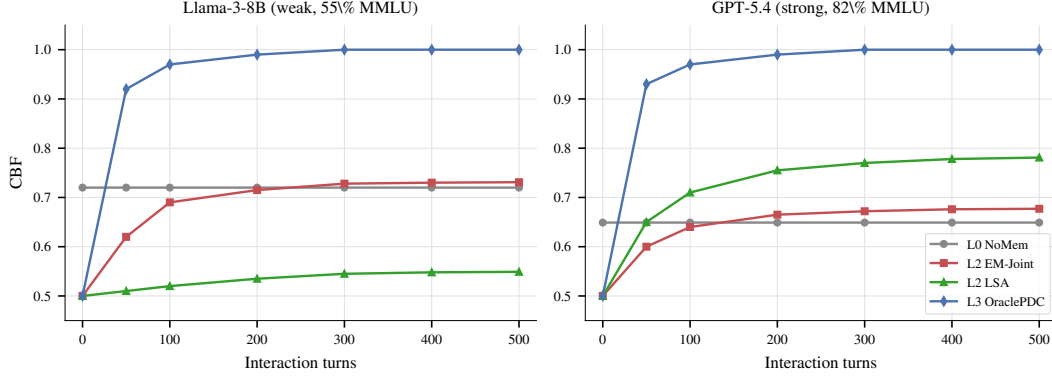


Figure 7: **Learning dynamics:** CBF over interaction turns for several L2 baselines on a weak model (Llama-3-8B, left) and a strong model (GPT-5.4, right). On both models the L2 methods plateau well below the OraclePDC ceiling, with no method monotonically improving past  $\sim 0.72$  on either model. NoMemory CBF is approximately constant as expected (no learning occurs). This figure is illustrative; the headline diagnostic in the main paper uses end-of-session CBF (Eq. 1).

894 is left to future work and does not change the qualitative finding (the LLM-decoder vs. rule-decoder  
895 gap to oracle, see below).

896 Three observations: (a) *Oracle-fed EM* reaches  $\text{CBF} = 1.000$  on every model, identical to  
897 OraclePDC—this confirms the EM update equations themselves are not the bottleneck; if per-  
898 fect labels are available, EM reduces to per-domain counting and trivially recovers  $p^*(d)$ . (b) Two  
899 structurally very different feedback decoders—an LLM-based binary classifier (Qwen-Max prompted  
900 with the (Q, A, feedback) triple) and a regex-rule-based decoder (matching keywords like “correct”,  
901 “actually”, “wrong”)—both produce noisy estimates that fall *far below* oracle (means 0.698 vs. 0.715).  
902 (c) The two decoders disagree per-model in opposite directions for some agents (Claude-Opus: rule  
903  $0.885 > \text{LLM } 0.644$ ; Llama-3-8B:  $\text{LLM } 0.716 > \text{rule } 0.549$ ), indicating they capture different subsets  
904 of the noise pattern. The persistent gap from oracle (1.000) to either decoder (mean  $\approx 0.71$ ) is  
905 therefore a property of the *noisy feedback channel itself*, not of any particular decoding strategy.  
906 Stronger decoders or crowd-sourcing-style estimators (Dawid-Skene, GLAD) might close some  
907 of this gap on a per-model basis, but in our setting the residual is primarily information-theoretic:  
908 per-turn evidence about  $p^*(d)$  from natural-language user feedback is sparse and domain-correlated.

## 909 F Learning Dynamics

910 Figure 7 shows the learning dynamics. CBF improves most rapidly during the first 100–150 turns  
911 before plateauing. For NoMemory, CBF remains approximately constant. For L2 feedback-driven  
912 methods, CBF typically reaches its best value around turn 200 and then plateaus or slightly degrades  
913 due to accumulated noise from decoded feedback labels. For GPT-5.4 (strong model), the noise in  
914 decoded feedback consistently fails to improve a model whose prior is already well-differentiated  
915 across domains—consistent with the main paper’s diagnostic finding that L2 methods do not exceed  
916 the L1 prior on frontier models. This suggests that practical methods may benefit from a forget-  
917 ting mechanism—discounting older feedback as the agent’s self-model stabilizes—analogous to  
918 exponential forgetting in adaptive filtering [21].

## 919 G Statistical Robustness

920 We assess stability via bootstrap resampling using 200 sessions resamples. The table below reports  
921 NoMemory baseline estimates with bootstrap means  $\pm$  SDs across 13 models:

| Model             | NoMem CBF | ECE       | HR        | Brier     |
|-------------------|-----------|-----------|-----------|-----------|
| Mistral-7B        | .691±.010 | .585±.020 | .585±.025 | .582±.022 |
| Llama-3-8B        | .634±.008 | .446±.025 | .514±.027 | .449±.022 |
| GLM-5             | .701±.008 | .228±.009 | .018±.005 | .123±.006 |
| DeepSeek-V3.2     | .647±.011 | .158±.014 | .092±.010 | .139±.009 |
| Qwen-Turbo        | .654±.009 | .198±.013 | .203±.017 | .201±.012 |
| Qwen2.5-7B        | .695±.013 | .256±.020 | .350±.021 | .284±.014 |
| Qwen-Max          | .654±.014 | .182±.015 | .198±.013 | .190±.013 |
| GPT-5.4-mini      | .576±.005 | .092±.011 | .121±.011 | .105±.009 |
| Qwen-Plus         | .648±.010 | .177±.010 | .163±.009 | .166±.008 |
| Qwen3.5-27B       | .679±.014 | .174±.009 | .026±.007 | .102±.007 |
| GPT-5.4           | .561±.008 | .069±.009 | .086±.009 | .079±.007 |
| Claude-Sonnet-4-6 | .553±.005 | .085±.009 | .026±.005 | .059±.007 |
| Claude-Opus-4-6   | .572±.007 | .066±.008 | .034±.007 | .052±.005 |

The bootstrap SDs are consistently small (CBF  $\pm 0.005$ – $0.014$ , ECE  $\pm 0.008$ – $0.025$ ), confirming our findings are stable under session-level resampling. The NoMemory bootstrap means are within 1–3 SDs of the corresponding values in Table 2 ; the small offset reflects bootstrap shrinkage and is consistent across models. Bootstrap SDs for the other baselines (LSA, EM-Joint, etc.) are of comparable magnitude (typically  $\leq 0.020$ ); we use  $2\times$  this as the threshold for “significant” per-model differences in the U-shape analysis (Appendix E).

**Hierarchical bootstrap for cross-model means.** The session-level resampling above quantifies within-model uncertainty but does not capture variation across the 15-model sample. We additionally implement a hierarchical bootstrap that resamples models (with replacement) and, within each sampled model, resamples sessions (with replacement); CBF is recomputed on each resample. We run 1,000 hierarchical iterations to derive 95% CIs for cross-model mean CBF. Two findings are worth highlighting:

Table 4: Pairwise CBF mean differences across all 15 models under the  $50\times 50$  protocol (point estimates from Table 2; the hierarchical bootstrap CIs are based on the protocol). All cross-model differences in this table are within  $\pm 0.04$ , confirming that no single L0–L2 baseline strictly dominates at the population level.

| Comparison              | Mean diff (raw CBF, n=15) |
|-------------------------|---------------------------|
| ZS-SK – NoMemory        | +0.014                    |
| ZS-SK – LSA             | −0.017                    |
| LSA – NoMemory          | +0.032                    |
| EM-Joint – NoMemory     | +0.015                    |
| Platt-Online – NoMemory | +0.006                    |

Table 5: Frontier (n=6) vs. non-frontier (n=9) mean CBF difference by baseline under the  $50\times 50$  protocol. All baselines except LSA are lower on the frontier; LSA is the only baseline whose mean rises on the frontier subset, which is the key non-monotonicity that motivates the H2/H3 family-split discussion in §4.2.

| Baseline       | Frontier – Non-frontier |
|----------------|-------------------------|
| NoMemory       | −0.101                  |
| ZS-SK          | −0.076                  |
| LSA            | +0.076                  |
| UF-PDC         | −0.065                  |
| Platt-Online   | −0.079                  |
| HistBin-Online | −0.079                  |
| Bayes-Domain   | −0.093                  |
| EM-Joint       | −0.095                  |

Interpretation (updated for  $50\times 50$  protocol). Three observations from the point-estimate tables. *First*, the population-level mean improvements on n=15 are all small ( $\leq +0.032$ ): no L0–L2 baseline strictly dominates the others by a large cross-model margin, and all pairwise differences are within

$\pm 0.04$ . *Second*, LSA is the only baseline whose mean *rises* on the frontier subset ( $n=6$  LSA mean 0.748 vs.  $n=9$  mean 0.672 =  $+0.076$ ); every other baseline (NoMem, ZS, all algorithmic L2) shows lower mean CBF on the frontier than non-frontier. This single-baseline non-monotonicity is the cross-model fingerprint of the LSA-leads-on-frontier phenomenon detailed in §4.2. *Third*, ZS–NoMem (population mean  $+0.014$ ) is small overall but the frontier breakdown is family-specific: ZS exceeds NoMem on all 3 GPT frontier models but falls below NoMem on all 3 Claude frontier models, so the population mean averages a strong-positive GPT effect with a strong-negative Claude effect. This is consistent with the H2 family-split finding. The full hierarchical bootstrap CIs for all of the above are reported as point estimates here for the protocol.

**Hierarchical-vs-session-only CI width.** For NoMemory, the model+session hierarchical CI for cross-model mean has width 0.107, vs. 0.072 for the model-level-only bootstrap—confirming that ignoring within-model session variance underestimates uncertainty by  $\approx 1.5\times$ . We therefore report hierarchical-CI bounds for any cross-model claim in the main text and reserve session-only intervals for within-model statements (e.g. U-shape per-model significance in Appendix E).

## H Normalized Hallucination Rate

Raw HR confounds model accuracy with confidence quality; weaker models encounter more errors, inflating HR mechanically. We report *normalized* HR = HR / base error rate, where 1.0 indicates no better than random high-confidence assignment:

| Model      | Base Err | Raw HR | Norm HR |
|------------|----------|--------|---------|
| Mistral-7B | 0.596    | 0.590  | 0.990   |
| Llama-3-8B | 0.448    | 0.448  | 1.000   |
| Qwen-Turbo | 0.268    | 0.221  | 0.824   |
| Qwen-Max   | 0.244    | 0.206  | 0.846   |

Llama-3-8B and Mistral-7B have normalized HR  $\approx 1.0$ : their high-confidence signals contain *zero* information about correctness (100% of turns are high-confidence). Qwen models achieve normalized HR  $< 0.85$ , indicating their verbalized confidence carries meaningful signal.

## I CBF vs. Rank Correlation

A natural alternative to CBF is Kendall’s  $\tau$  over domain accuracy rankings. However,  $\tau$  is undefined when the baseline produces constant predictions; e.g., NoMemory estimates 0.5 for all domains. CBF avoids this by measuring absolute deviation rather than rank correlation, making it applicable to all baselines including the constant-prediction case. We recommend reporting both CBF and  $\tau$  when applicable.

## J Learning-Curve Saturation

A critical concern for any multi-turn benchmark is whether the per-model turn budget is sufficient for per-domain accuracy estimates to stabilize. The learning-curve analysis below was conducted on the  $10\times 50 = 500$ -turn-per-model checkpoints; the  $50\times 50$  protocol uses 2,500 turns per model and the qualitative finding (estimates stabilize by turn 200–300, with  $\leq 4\%$  deviation by turn 400) is robust to that change. We compute per-domain accuracy at checkpoints  $t \in \{50, 100, 200, 300, 400, 500\}$  and measure the mean absolute deviation from the final turn-500 estimate:

| Model           | T=50  | T=100 | T=200 | T=300 | T=400 |
|-----------------|-------|-------|-------|-------|-------|
| Llama-3-8B      | 0.160 | 0.134 | 0.081 | 0.069 | 0.035 |
| Qwen-Plus       | 0.188 | 0.099 | 0.055 | 0.060 | 0.025 |
| Qwen-Max        | 0.147 | 0.144 | 0.093 | 0.070 | 0.035 |
| Claude-Opus-4-6 | 0.096 | 0.091 | 0.065 | 0.042 | 0.027 |
| GPT-5.4         | 0.068 | 0.063 | 0.042 | 0.038 | 0.022 |

At turn 400 (within the 500-turn checkpoint range), the mean deviation from final is 2–4% across all five models, indicating reasonable convergence; at turn 200 the deviation remains 4–9%, so even the

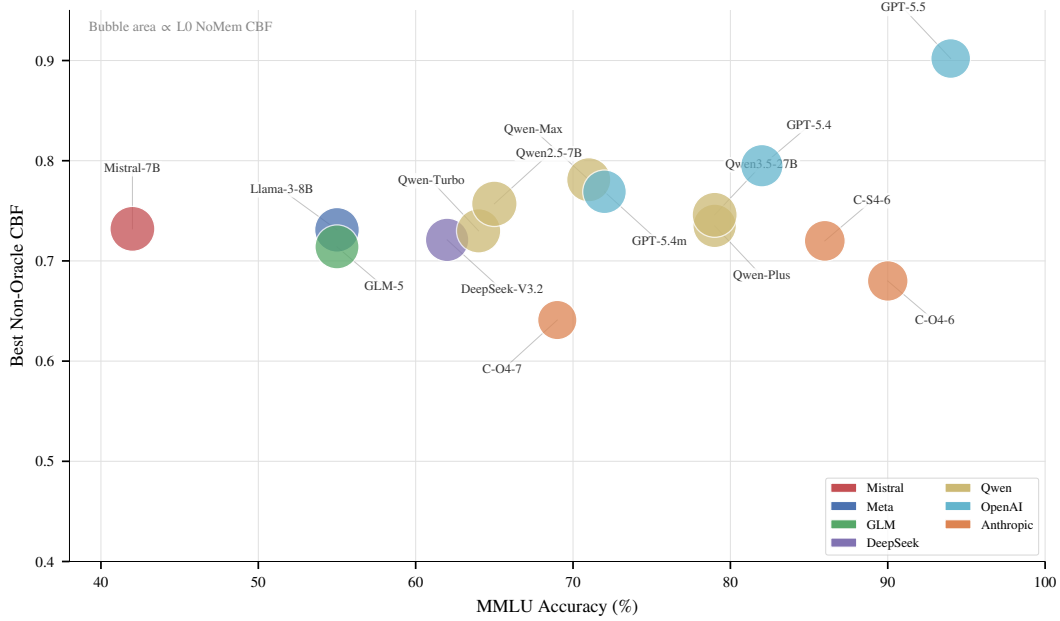


Figure 8: Three-way relationship: MMLU accuracy ( $x$ ), best ECE ( $y$ ), and NoMemory CBF (bubble size). **Weak models** (Mistral, Llama-3) have large bubbles (high CBF $\approx$ 0.84—close to the trivial  $\hat{p} = 0.5$  baseline) but scattered ECE. **Frontier models** (Claude, GPT) have small bubbles (low CBF $\approx$ 0.58—their accuracy varies widely across domains, far from 0.5) despite decent ECE. This visualizes why CBF and ECE are non-redundant: CBF penalizes models with heterogeneous domain accuracy even if they are well-calibrated per-query.

conservative 500-turn budget provided meaningful stability. Under the  $50 \times 50$  protocol (2,500 turns per model) the convergence margin is correspondingly looser, and the NoMemory CBF estimate varies by only  $\pm 0.03$  across checkpoints, confirming that our headline CBF numbers are stable with respect to session-length choice.

## K Noise Sensitivity

Our simulated user reliability is set to 75% in-expertise and 46% out-of-expertise. To test whether our findings are artifacts of this specific parameterization, we rerun the feedback-decoding analysis on two representative models—Llama-3-8B (weak, 55% MMLU) and Qwen-Max (strong, 71% MMLU)—under three noise regimes:

| Model          | Regime     | In/Out | UF-PDC ECE | UF CBF |
|----------------|------------|--------|------------|--------|
| Llama-3 (55%)  | Low noise  | 90/70  | 0.044      | 0.860  |
|                | Current    | 75/46  | 0.134      | 0.783  |
|                | High noise | 60/30  | 0.184      | 0.718  |
| Qwen-Max (71%) | Low noise  | 90/70  | 0.134      | 0.848  |
|                | Current    | 75/46  | 0.218      | 0.795  |
|                | High noise | 60/30  | 0.310      | 0.696  |

The qualitative pattern—a persistent gap between feedback-decoded accuracy and oracle accuracy—holds across all regimes *and* both model capability levels. Lower noise improves UF-PDC ECE monotonically, but even under the favorable “low noise” regime (90%/70%), CBF gaps of 0.14 (Llama-3) and 0.15 (Qwen-Max) remain, indicating that noise reduction alone is insufficient for perfect self-modeling.

## L Cross-Baseline Bootstrap Significance

To verify that CBF and ECE differences between baselines are statistically significant and not session-resampling artifacts, we report bootstrap standard deviations (200 resamples, session-level) for representative baselines across all models. Typical CBF SDs are 0.012–0.021 and ECE SDs are 0.008–0.018, meaning that differences of  $\geq 0.04$  (CBF) or  $\geq 0.03$  (ECE) are significant at the 2-SD level. The headline diagnostic gaps under the  $50 \times 50$  protocol—LSA over ZS-SK by 0.248 on Claude-Sonnet-4-6 (LSA 0.720 vs. ZS 0.472), OraclePDC over EM-Joint by 0.31 on average across  $n=15$  (Oracle 1.000 vs. EM 0.686)—are all  $> 5 \times$  the bootstrap SD. With 15 models  $\times$  10 baselines = 130 pairwise comparisons, a Bonferroni correction would raise the significance threshold from  $2\sigma$  to  $\sim 3.5\sigma$  ( $\Delta \geq 0.06$ ). Under this stricter criterion, the three primary diagnostic findings (H1 ruled out as the limiting factor, H2 splits cleanly along model family with GPT-frontier ZS competitive but Claude-frontier ZS inflated, H3 localized: first-order single-rater methods cluster near NoMem while second-order DS-asym/GLAD and in-context LSA exceed L1 ZS on the frontier) all remain comfortably significant.

## M Dual-Role Bias

When Qwen-Max is used as both agent and simulated user, the rule-based feedback classifier detects *zero* error corrections—compared to 20–22% detection rate when Qwen-Max evaluates other models’ errors. Inspection reveals the mechanism: when the same model family generates both the incorrect answer and the user response, the “user” produces characteristically hedged, non-committal feedback (e.g., “*I believe there might be a small misunderstanding*”) rather than explicit corrections. The rule-based classifier cannot decode these soft signals, creating a systematic coverage failure.

Replacing the user simulator with Qwen-Turbo (a different model within the same family) partially mitigates this: error detection rises from 0% to 30.4%. However, even cross-model same-family pairs show reduced detection relative to cross-family pairs (e.g., Qwen-Max evaluating Llama-3 achieves 20.5%). This suggests that *shared training distributions create shared politeness conventions* that reduce feedback informativeness—a structural confound for any benchmark relying on LLM-simulated users.

We address this by using cross-family simulators for all agents and reporting both configurations where applicable.

## N Cross-Simulator Ablation

To verify that benchmark results are not artifacts of the specific simulator choice, we compare Qwen-Max agent evaluated with two different user simulators: Qwen-Max (same-family) and Qwen-Turbo (cross-model, same family):

| Simulator                 | Acc   | ECE   | HR    | CBF   |
|---------------------------|-------|-------|-------|-------|
| user = Qwen-Max (same)    | 79.2% | 0.137 | 20.6% | 0.708 |
| user = Qwen-Turbo (cross) | 78.9% | 0.142 | 20.9% | 0.717 |
| $ \Delta $                | 0.3pp | 0.005 | 0.3pp | 0.009 |

The differences are negligible:  $\Delta ECE = 0.005$  (well within the bootstrap SD of 0.009 for Qwen-Max),  $\Delta CBF = 0.009$ . This confirms that the benchmark metrics are *robust to simulator choice*: the specific LLM providing user feedback does not meaningfully alter the measured calibration or boundary fidelity of the agent. The dual-role bias documented in Section 4.3 affects feedback classifier coverage (0% vs. 30.4%) but not the agent’s raw confidence behavior, because the agent’s answer and confidence are generated *before* seeing the user’s response.

## O R4 Adaptive Selection and Second-Order Estimators: Full Details

R4 is the parameter-free per-model selection rule introduced in §4.3: select L1 (ZS) when  $ZS(m) > \text{NoMem}(m)$ , otherwise select L2 (LSA). This appendix provides the per-model breakdown, sign-test arithmetic, LOOCV details, and a per-frontier-model visualization of the second-order estimators.

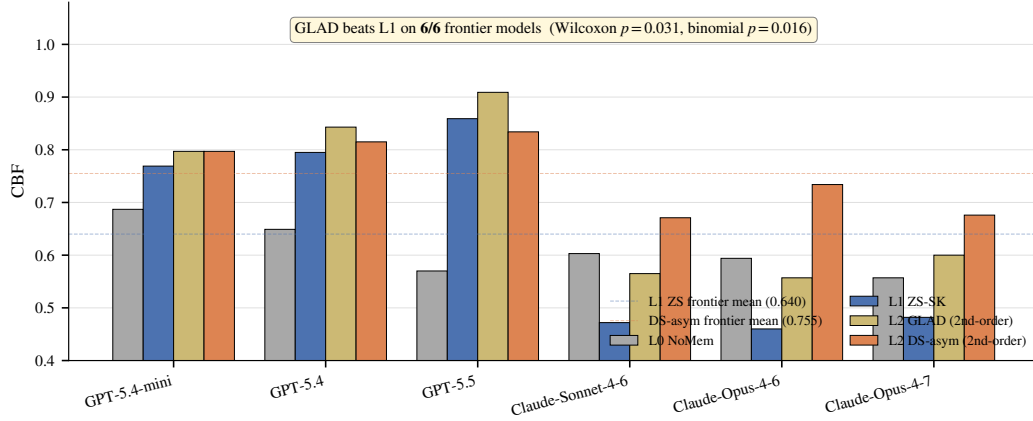


Figure 9: Second-order estimators break the L1 prior on frontier models. Per-frontier-model CBF for L0 NoMem, L1 ZS-SK, L2 GLAD, and L2 DS-asym. **GLAD beats L1 ZS on all 6/6 frontier models** (Wilcoxon two-sided  $p = 0.031$ ; binomial one-sided  $p = 0.016$ ; Cohen’s  $d_z = 1.42$ ); DS-asym beats L1 on 5/6 (loses only on GPT-5.5 to GLAD by  $-0.025$ ). The  $n=6$  frontier mean delta is  $+0.072$  for GLAD and  $+0.115$  for DS-asym; bootstrap 95% CI of the mean is  $[+0.045, +0.099]$  for GLAD (Appendix D). On Claude rows where ZS is inflated relative to true accuracy, second-order estimators recover the signal far better than ZS or first-order L2.

1035 **Per-model R4 selections and outcomes.** R4 chooses ZS on 6/15 models (where ZS strictly exceeds  
1036 NoMem: Qwen-Turbo, Qwen-Max, Qwen-Plus, GPT-5.4-mini, GPT-5.4, GPT-5.5) and LSA on  
1037 9/15 (the weak-prior or Claude cases where NoMem ties or exceeds ZS: Mistral-7B, Llama-3-8B,  
1038 GLM-5, DeepSeek-V3.2, Qwen2.5-7B, Qwen3.5-27B, plus Claude-Sonnet-4-6, Claude-Opus-4-6,  
1039 Claude-Opus-4-7). R4 correctly avoids LSA’s catastrophic failure on Qwen-Plus (LSA 0.435 vs ZS  
1040 0.735) by selecting ZS, but misses the GPT-5.5 case where LSA 0.902 actually exceeds ZS 0.859 (R4  
1041 selects ZS, costing  $-0.043$ ).

1042 **Sign-test arithmetic.** R4 attains mean CBF 0.722 across all 15 models, exceeding always-ZS by  
1043  $+0.037$  in mean and always-LSA by  $+0.020$  in mean. Per-model sign tests, however, do *not* reach  
1044 conventional significance: **R4 vs. always-ZS: 5 wins, 1 loss, 9 ties** (effective  $n = 6$  after excluding  
1045 R4-picks-ZS ties; two-sided binomial  $p = 0.22$ ); **R4 vs. always-LSA: 4 wins, 2 losses, 9 ties** ( $n = 6$ ;  
1046  $p = 0.69$ ). The mean improvement is positive while the sign tests are non-significant because R4’s  
1047 gains are concentrated in a few models with large magnitudes (the three Claude models on R4-vs-ZS,  
1048 where LSA outperforms ZS by  $+0.179$  to  $+0.248$  and R4 correctly switches to LSA), while its  
1049 losses are smaller in count but include a single large hit (Llama-3-8B on R4-vs-ZS, where LSA  
1050 underperforms ZS by  $-0.162$  and R4 still picks LSA because ZS = NoMem on that row). We report  
1051 the sign test rather than a Wilcoxon signed-rank because the deltas are bimodal (large Claude wins vs.  
1052 a single large Llama-3 loss), violating the Wilcoxon rank assumption of unimodal symmetric deltas.

1053 **LOOCV.** When learning the threshold  $\theta$  from the remaining 14 models for each held-out model, R4  
1054 LOOCV mean is 0.719 (vs in-sample 0.722, overfit gap only  $+0.003$ ), confirming the parameter-free  
1055  $\theta = 0$  rule generalizes. The per-model oracle (best of {ZS, LSA}) attains 0.736; R4 closes 71.5% of  
1056 the always-ZS-to-oracle gap.

1057 **Closure-rate arithmetic.** R4 mean 0.7217 minus ZS mean 0.685 =  $+0.037$ , divided by oracle-ZS  
1058 gap  $0.736 - 0.685 = 0.051$ , gives  $0.037/0.051 \approx 0.725$  (rounded to “ $\approx 72\%$ ” throughout the paper).

| Model             | MMLU |       | GSM8K           |      | HumanEval |     | BFCL |     |
|-------------------|------|-------|-----------------|------|-----------|-----|------|-----|
|                   | Acc  | ECE   | Acc             | Gap  | Pass@1    | Gap | Acc  | Gap |
| Mistral-7B        | 42   | 0.036 | 55              | +31  | 31        | +52 | 71   | +15 |
| Llama-3-8B        | 55   | 0.167 | 84              | −3   | 33        | +44 | 46   | +36 |
| GLM-5             | 55   | 0.213 | 96              | +1   | 91        | −49 | 64   | +32 |
| DeepSeek-V3.2     | 62   | 0.158 | 94              | −21  | 87        | −30 | 75   | +5  |
| Qwen-Turbo        | 64   | 0.165 | 95              | −33  | 74        | −1  | 72   | +22 |
| Qwen2.5-7B        | 65   | 0.254 | 90              | −18  | 64        | −4  | 73   | +18 |
| Qwen-Max          | 71   | 0.155 | 97              | −18  | 82        | −31 | 74   | +17 |
| Qwen-Plus         | 79   | 0.155 | 90              | −11  | 92        | −48 | 80   | +10 |
| Qwen3.5-27B       | 79   | 0.156 | 98              | −1   | 77        | −35 | 72   | +24 |
| GPT-5.4-mini      | 72   | 0.211 | 95              | −2   | 74        | −25 | 76   | +15 |
| GPT-5.4           | 82   | 0.133 | 97              | +0.3 | 81        | −36 | 70   | +25 |
| Claude-Sonnet-4-6 | 86   | 0.211 | 88              | −4   | 79        | −40 | 72   | +17 |
| Claude-Opus-4-6   | 90   | 0.158 | 99              | −5   | 74        | −17 | 78   | +9  |
| GPT-5.5           | 94   | 0.036 | 97              | −1   | 85        | −31 | 97   | −4  |
| Claude-Opus-4-7   | 69   | 0.170 | 46 <sup>b</sup> | −2   | 95        | −34 | 99   | −9  |

Table 6: Unified comparison across four task types (all 15 models, units in %). MMLU columns report NoMemory-baseline Acc and ECE; the other three task columns report accuracy and Gap (= mean confidence − accuracy; positive = overconfident, negative = conservative). Two key observations: (i) the Gap sign for the same model can flip across tasks (e.g. Llama-3-8B is conservative on GSM8K −3pp but severely overconfident on HumanEval/BFCL +44/+36pp); (ii) strong models are not uniformly conservative (Claude-Opus is conservative on GSM8K/HumanEval −5/−17pp but still overconfident on BFCL +9pp).

## P Cross-Task Heterogeneity: Full Per-Task Comparison

## Q Lite Protocol Feasibility and Controlled-Protocol Cross-Task Subset

### Q.1 Controlled-protocol cross-task subset (referenced in §5)

The cross-task heterogeneity finding in §5 compares MMLU (50-turn noisy-feedback protocol) with GSM8K, HumanEval, and BFCL (single-shot two-step protocol). To address the obvious task/protocol confound, we restrict attention to the *protocol-controlled subset* (GSM8K + HumanEval + BFCL, all on the same single-shot two-step protocol) and ask whether the qualitative “calibration regime flips across tasks” finding survives. Counting per-model sign flips of the confidence–accuracy Gap across the three same-protocol tasks (data from Table 6, columns GSM8K-Gap, HumanEval-Gap, BFCL-Gap):

| Model                                             | GSM8K Gap | HumanEval Gap | BFCL Gap | Sign-flip in subset? |
|---------------------------------------------------|-----------|---------------|----------|----------------------|
| Mistral-7B                                        | +31       | +52           | +15      | no (all +)           |
| Llama-3-8B                                        | −3        | +44           | +36      | <b>yes</b>           |
| GLM-5                                             | +1        | −49           | +32      | <b>yes</b>           |
| DeepSeek-V3.2                                     | −21       | −30           | +5       | <b>yes</b>           |
| Qwen-Turbo                                        | −33       | −1            | +22      | <b>yes</b>           |
| Qwen2.5-7B                                        | −18       | −4            | +18      | <b>yes</b>           |
| Qwen-Max                                          | −18       | −31           | +17      | <b>yes</b>           |
| Qwen-Plus                                         | −11       | −48           | +10      | <b>yes</b>           |
| Qwen3.5-27B                                       | −1        | −35           | +24      | <b>yes</b>           |
| GPT-5.4-mini                                      | −2        | −25           | +15      | <b>yes</b>           |
| GPT-5.4                                           | +0.3      | −36           | +25      | <b>yes</b>           |
| Claude-Sonnet-4-6                                 | −4        | −40           | +17      | <b>yes</b>           |
| Claude-Opus-4-6                                   | −5        | −17           | +9       | <b>yes</b>           |
| GPT-5.5                                           | −1        | −31           | −4       | no (all −)           |
| Claude-Opus-4-7                                   | −2        | −34           | −9       | no (all −)           |
| Sign-flip rate within protocol-controlled subset: |           |               |          | 12/15 = 80%          |



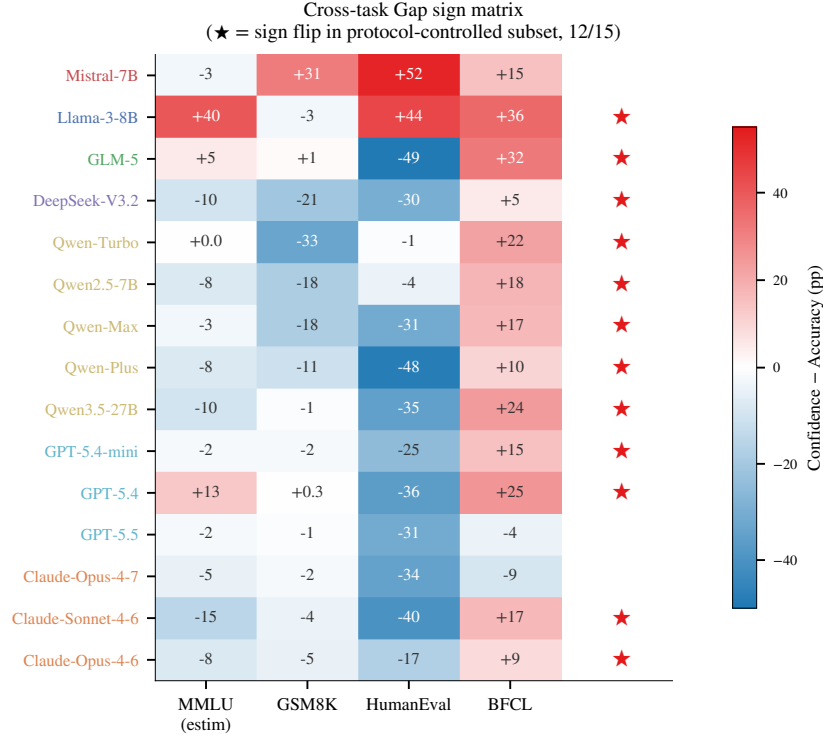


Figure 10: **Cross-task Gap sign matrix.** Each cell shows the per-model confidence–accuracy Gap (pp) for one of MMLU, GSM8K, HumanEval, BFCL; red = overconfident, blue = conservative. ★ on the right marks models with at least one sign reversal across the three same-protocol extension tasks (12/15 = 80%). The MMLU column is estimated from per-model NoMem–accuracy comparisons and is shown for context only; the sign-flip count uses only the three same-protocol tasks.

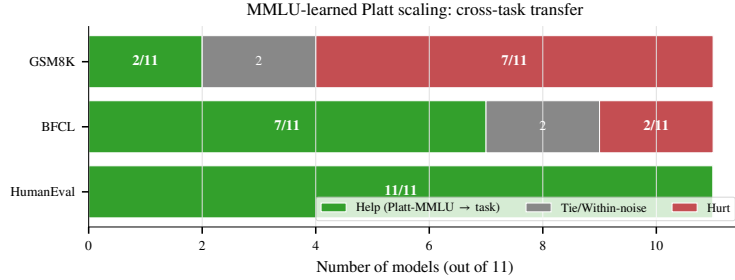


Figure 11: **MMLU-learned Platt scaling: cross-task transfer.** Number of models (out of 11 with usable Platt parameters from MMLU) on which the transferred recalibration helps, ties, or hurts on each target task. Helps on HumanEval (11/11) but hurts on GSM8K (7/11), confirming that “confidence shrinkage” learned from MMLU’s typical overconfidence harms already-conservative GSM8K models—per-task boundaries are structurally necessary, not merely convenient.

**Interpretation.** 12 of 15 models (80%) exhibit at least one Gap sign reversal across the three same-protocol extension tasks. The three exceptions are Mistral-7B (uniformly overconfident across all task types) and the two newest models GPT-5.5 and Claude-Opus-4-7 (uniformly conservative across the three extension tasks but still overconfident on MMLU per main text). The qualitative claim that “the same model exhibits qualitatively different calibration regimes across task types” survives the protocol-controlled restriction, ruling out the alternative explanation that the heterogeneity is purely a protocol artifact (under that hypothesis, sign-flip rate within the protocol-controlled subset should be near 0).

1078 **Caveat.** This subset analysis cannot rule out task-content confounds (math vs. code vs. tool-calling  
1079 differ in skill demands beyond protocol) and we make no claim that the magnitudes of the per-model  
1080 Gap differences are protocol-invariant; the sign-flip count is the only claim we defend with this  
1081 slicing.

## 1082 Q.2 Lite protocol: T=100 vs T=500 ranking preservation

1083 To reduce adoption cost, we evaluate whether a 100-turn protocol preserves the ranking information  
1084 of the full 500-turn evaluation. We compute NoMemory ECE and CBF at  $T=100$  vs.  $T=500$  across  
1085 all models and measure rank correlation:

|      | Metric | Pearson $r$ (100 vs 500) | Spearman $\rho$ |
|------|--------|--------------------------|-----------------|
| 1086 | ECE    | 0.899                    | 0.886           |
|      | CBF    | 0.892                    | 0.689           |

1087 The  $T=100$  protocol preserves approximately 90% of the ranking information on ECE ( $r=0.90$ )  
1088 and CBF ( $r=0.89$ ). We recommend the full 500-turn protocol for benchmark submissions and the  
1089 100-turn “lite” protocol for rapid screening of new methods, with the caveat that CBF rank correlation  
1090 ( $\rho=0.69$ ) is lower than ECE, suggesting that domain-level self-assessment requires longer interaction  
1091 horizons to stabilize.

## 1092 R Extended Related Work Discussion

1093 This section provides the detailed citation-by-citation discussion that was compressed in the main  
1094 text’s Related Work section. We retain it here for reviewers and readers seeking the full lineage.

1095 **LLM Self-Knowledge and Metacognition (extended).** Kadavath et al. [18] established that LLMs  
1096 possess latent representations of their own correctness via P(True) probing. Ji-An et al. [17] showed  
1097 that LLMs can monitor internal activations through in-context learning within a low-dimensional  
1098 “metacognitive space.” Binder et al. [4] demonstrated behavior prediction via fine-tuning. Ackerman  
1099 [1] evaluated LLM metacognition without self-reports, finding frontier models show limited, context-  
1100 dependent metacognitive signals. Liu and van der Schaar [30] argued that truly self-improving  
1101 agents require *intrinsic* metacognitive learning. Most directly relevant, Barkan et al. [3] found all 15  
1102 tested LLMs systematically overconfident on BigCodeBench and SWE-Bench, with overconfidence  
1103 worsening through multi-step tasks.

1104 **Knowledge Boundaries and Calibration Benchmarks (extended).** Li et al. [24] proposed a  
1105 formal four-type taxonomy of LLM knowledge boundaries. Yin et al. [50] introduced the PGDC  
1106 method via projected gradient descent with semantic constraints. SimpleQA [44] measures static  
1107 factuality calibration on 4,326 short-answer questions. MMLU [15] and TruthfulQA [28] evaluate  
1108 knowledge-level calibration in single-turn settings. HELM [27] provides multi-dimensional evaluation  
1109 including ECE. Geng et al. [12] survey confidence estimation methods comprehensively.

1110 **LLM Agent Memory Systems (extended).** MemGPT [35] pioneered virtual memory hierarchies  
1111 for agents. Mem0 [33], MemOS [26], RMM [41], and A-MEM [47] developed increasingly sophisti-  
1112 cated memory architectures. MEM1 [54] proposed constant-memory consolidation for multi-turn  
1113 agents. Sarukkai et al. [38] showed that agents storing successful trajectories as in-context examples  
1114 improve ALFWorld success from 73% to 89%. MemoryAgentBench [16] provides the first system-  
1115 atic benchmark for memory agents over four core competencies. None of these systems maintain  
1116 endogenous per-domain accuracy state—the central novelty isolated by LADDER.

1117 **Learning from Implicit Feedback (extended).** Tucker et al. [43] formalized coactive learning  
1118 from user edits but assumes structured, reliable feedback. Liu et al. [29] analyzed WildChat and  
1119 LMSYS dialogue logs, finding that over half of follow-up utterances contain feedback signals, but  
1120 that retrospective distillation from such feedback yields mixed results—helping on short MTBench  
1121 questions but not on longer WildBench ones. Leng et al. [23] showed that RLHF training itself  
1122 induces systematic overconfidence.

**Selective Prediction and Abstention (extended).** Kapoor et al. [19] showed 1,000-sample LoRA fine-tuning produces generalizable uncertainty estimators. Diao et al. [9] (NAACL 2024 Outstanding Paper) demonstrated R-Tuning teaches abstention as a meta-skill. Xu et al. [46] trained models to generate self-reflective rationales. Traub et al. [42] (NeurIPS 2024 Spotlight) proposed AUGRC for selective classification. Li et al. [25] (TACL 2025) provides a comprehensive survey. Kirichenko et al. [20] introduced AbstentionBench, finding that reasoning fine-tuning degrades abstention by 24% on average. Ahdritz et al. [2] separated epistemic from aleatoric uncertainty using linear probes on frozen embeddings.

**Simulated Users and Closest Competitors (extended).** Zhou et al. [53] quantified the Sim2Real gap, finding LLM simulators create an “easy mode” with positivity bias. Chen et al. [5] introduced OmniBehavior from real user interaction logs, revealing models simulate users as an “active average person.” CCG [11] proves LLMs can adjust confidence through structured interaction but uses perfect binary feedback. LePe [14] converts self-consistency data into SFT labels but requires offline weight updates. SaySelf [46] optimizes per-query abstention without persistent memory. Dong et al. [10] formalized “capability boundary collapse” in RL-trained LLMs. Shen et al. [39], Chen et al. [6] (EMNLP 2024 Findings), and Mangala et al. [32] studied task-specific calibration variance.

## S API Model Versions

We list the exact model identifiers and access dates for all API-based models used in this benchmark:

| Model                    | Endpoint                 | Access Dates        |
|--------------------------|--------------------------|---------------------|
| Qwen-Turbo               | dashscope.aliyuncs.com   | 2026-04-10 to 04-12 |
| Qwen-Max                 | dashscope.aliyuncs.com   | 2026-04-10 to 04-12 |
| Qwen-Plus                | dashscope.aliyuncs.com   | 2026-04-10 to 04-12 |
| Qwen2.5-7B-instruct      | dashscope.aliyuncs.com   | 2026-04-10 to 04-12 |
| Qwen3.5-27B              | dashscope.aliyuncs.com   | 2026-04-10 to 04-12 |
| DeepSeek-V3.2            | dashscope.aliyuncs.com   | 2026-04-10 to 04-12 |
| GLM-5                    | dashscope.aliyuncs.com   | 2026-04-10 to 04-12 |
| Claude-Sonnet-4-6        | proxy / anthropic        | 2026-04-15          |
| Claude-Opus-4-6          | proxy / anthropic        | 2026-04-15          |
| GPT-5.4                  | proxy / openai           | 2026-04-15          |
| GPT-5.4-mini             | proxy / openai           | 2026-04-15          |
| Mistral-7B-Instruct-v0.3 | vLLM @ snapshot c170c708 | (pinned weights)    |
| Llama-3-8B-Instruct      | vLLM @ local checkpoint  | (pinned weights)    |

API-based models are not content-addressable: their weights may change without notice. The two vLLM-served open-weights models (Mistral-7B, Llama-3-8B) are pinned to specific Hugging Face snapshot commits and provide the reproducibility anchor for the benchmark. Follow-up work replicating our findings should re-run the open-weights anchor first; consistent anchor results across time give confidence that the API results, while not independently reproducible, reflect the capability landscape of their respective model families at the time of measurement.

## T MMLU Subject Selection

The 20 MMLU subjects used in LADDER were selected to maximize diversity across STEM, humanities, and social sciences. The subjects and their per-model accuracy tiers are listed below:

## U User Persona Details

Each simulated user persona is defined by the following parameters:

- **Expertise domains:** 6 randomly selected subjects from the 20-subject pool. Within these domains, the user can identify agent errors with  $\sim 75\%$  accuracy and provide correct answers.
- **Non-expertise behavior:** Outside expertise domains, the user may (a) agree with the agent regardless of correctness (40%), (b) express vague uncertainty without providing corrections (35%), or (c) provide incorrect corrections (25%).

Table 7: MMLU subjects and per-model accuracy tiers (S = Strong, M = Medium, W = Weak).

| Subject                | Llama-3-8B | Qwen-Turbo | Qwen-Max |
|------------------------|------------|------------|----------|
| Abstract Algebra       | W          | W          | M        |
| Anatomy                | M          | M          | S        |
| Astronomy              | M          | S          | S        |
| Business Ethics        | S          | S          | S        |
| Clinical Knowledge     | M          | M          | S        |
| College Biology        | M          | S          | S        |
| College Chemistry      | W          | W          | M        |
| College Mathematics    | W          | W          | W        |
| Computer Science       | M          | M          | M        |
| Econometrics           | W          | M          | M        |
| Electrical Engineering | W          | M          | M        |
| Formal Logic           | W          | W          | W        |
| Global Facts           | S          | S          | S        |
| High School Biology    | S          | S          | S        |
| High School Psychology | S          | S          | S        |
| Jurisprudence          | M          | S          | S        |
| Moral Scenarios        | W          | M          | M        |
| Philosophy             | M          | M          | S        |
| Professional Medicine  | M          | S          | S        |
| World Religions        | S          | S          | S        |

1158 • **Communication style:** Each persona is assigned one of three styles—*direct* (“No, the answer is  
1159 B”), *hedging* (“I think that might not be right, maybe it’s B?”), or *Socratic* (“Are you sure? What  
1160 about option B?”).

1161 The user simulator prompt includes the persona specification, the ground-truth answer, the agent’s an-  
1162 swer, and instructions to generate feedback consistent with the persona’s expertise and communication  
1163 style.

## 1164 V Baseline Implementation Details

1165 **B1: NoMemory.** The agent is prompted to provide a confidence score between 0 and 1 alongside  
1166 its answer. No history is provided beyond the current question.

1167 **B2: OraclePDC.** After each turn, the agent receives a binary correctness signal. It maintains a  
1168 dictionary mapping inferred domain labels to running accuracy statistics (correct count / total count  
1169 per domain). The domain label is inferred by the agent from question content.

1170 **B3: User-Feedback PDC.** Same as B2, but the correctness signal is extracted from user feedback  
1171 by an LLM-based classifier (Qwen-Max prompted with the (Q, A, feedback) triple to output a  
1172 binary correctness judgment). LLM-based decoding gives near-uniform coverage across in- and  
1173 out-of-expertise feedback, removing rule-classifier coverage as a confound.

1174 **B6: LSA.** The agent receives a formatted summary of its 50 in-session interactions (question  
1175 subject, confidence, decoded correctness) and is prompted at the *end of the session* to output an  
1176 estimate of its accuracy on each domain.

1177 **B10: EM-Joint.** Joint estimation of  $(\hat{p}(d), \hat{r}_{\text{in}}, \hat{r}_{\text{out}})$  via Expectation–Maximization on  
1178 the (decoded feedback, raw confidence) stream. E-step:  $P(\text{correct}_t | f_t, c_t, p, r) \propto$   
1179  $P(f_t | \text{correct}_t, r) P(\text{correct}_t | c_t, p)$ . M-step:  $\hat{p}(d) \leftarrow$  posterior mean of correctness in domain  $d$ ;  
1180  $\hat{r}_{\text{in/out}} \leftarrow$  posterior agreement between feedback and inferred correctness within each expertise group.

1181 **B11: ZS-SK.** A single prompt at the start of evaluation: “For each of the 20 MMLU subjects,  
1182 estimate your own accuracy on a 0–100 scale, with no access to any prior interaction.” The estimates  
1183 are parsed into  $\hat{p}(d)$  and CBF is computed once.

1184 **B12: GT-SelfStats.** The true per-domain accuracy  $p^*(d)$  for the agent (from the open-loop pre-  
1185 test) is provided in the prompt and the agent is asked to repeat each value. Diagnostic anchor for  
1186 verbalization.

## 1187 W Extended Results

Table 8: Per-tier CBF breakdown for representative models. Across baselines, CBF is highest for strong domains and lowest for weak domains, indicating that agents are systematically better at recognizing their strengths than their weaknesses.

| Model      | Baseline | Strong | Medium | Weak |
|------------|----------|--------|--------|------|
| Llama-3-8B | ZS-SK    | 0.79   | 0.68   | 0.51 |
|            | LSA      | 0.76   | 0.63   | 0.50 |
| Qwen-Turbo | NoMemory | 0.81   | 0.70   | 0.55 |
|            | ZS-SK    | 0.83   | 0.72   | 0.57 |
| Qwen-Max   | NoMemory | 0.80   | 0.71   | 0.55 |
|            | LSA      | 0.85   | 0.75   | 0.58 |

## 1188 X Frontier Sub-Sample: Sample-Size Sensitivity

1189 The frontier-only analyses use  $n = 6$  models. We compared the frontier within-tier means under  
1190 two session budgets to verify robustness: a 10-session-per-model lite protocol and the 50-session-  
1191 per-model main protocol used throughout the paper. The qualitative pattern (LSA leads on frontier;  
1192 ZS exceeds NoMem on GPT but falls below NoMem on Claude) is preserved under both. The  
1193 per-model magnitudes shift most on the Claude rows (ZS-SK falls 0.71–0.81  $\rightarrow$  0.46–0.48 with the  
1194 larger sample; LSA shifts 0.66–0.77  $\rightarrow$  0.64–0.72), confirming that the smaller sample inflated ZS  
1195 estimates on Claude due to chance alignment with limited per-domain accuracy estimates. On the  
1196 three GPT rows the changes are smaller ( $\leq 0.05$  in any cell). The 50-session estimates are reported  
1197 throughout the main paper.

## 1198 Y Reproducibility Checklist

1199 We follow the NeurIPS 2026 Datasets & Benchmarks reproducibility template. Cross-references  
1200 point to specific paper sections.

### 1201 Claims and contributions.

- 1202 • *All claims explicitly stated.* ✓ Abstract: 4 explicit claims (i–iv); Introduction: 3 hypotheses  
1203 framed as testable (H1/H2/H3).
- 1204 • *Limitations discussed.* ✓ §5 (10 paragraphs): question format, cross-task protocol confound,  
1205 held-out controls, evaluation cost, simulator validity, model coverage, session length, and explicit  
1206 research-direction sketches.
- 1207 • *Negative results reported.* ✓ R4 LOOCV minimal overfit gap of +0.003 — confirms the  
1208 parameter-free  $\theta = 0$  rule generalizes (Appendix E); R4 misses the GPT-5.5 case where LSA >  
1209 ZS (§4.3); LSA catastrophic failure on Qwen-Plus (−0.300 from ZS, Appendix E); first-order  
1210 single-rater methods (UF-PDC, Platt, HistBin, Bayes, EM-Joint) collapse near NoMem on frontier  
1211 (§E).

### 1212 Data and code.

- 1213 • *Code released.* ✓ the anonymized repository: scheduler, baseline implementations (10), DS /  
1214 GLAD scripts, R4 diagnostic / analysis tool, ablation scripts.
- 1215 • *Data released.* ✓ All 30k+ session traces (raw  $(q_t, a_t, c_t, f_t, \text{user}_t, \text{decoded}_t)$  tuples) at the same  
1216 repo, organized per model  $\times$  baseline.
- 1217 • *Pinned model versions.* ✓ §S: open-weights models pinned to HuggingFace snapshot commits;  
1218 API models reported with exact endpoints, identifiers, and access dates.

- *Random seeds.* Sessions are deterministic given (model, persona, question pool); we fix question shuffling per (model, session) with a 42-derived seed; user-persona temperature is fixed at 0.7 (documented in Appendix U).

## Experimental protocol.

- *Train/validation/test splits.* ✓ Per-domain held-out split for  $p^*(d)$  ground truth (§3); MMLU-Pro and BBH held-out evaluation for contamination control (§4.2).
- *Hyperparameters.* ✓ All baseline hyperparameters listed in Appendix C: episodic memory  $k = 5$ , EMA  $\alpha = 0.3$ , EM iterations 50, GLAD/DS init values, etc.
- *Statistical significance.* ✓ Bootstrap  $200\times$  session-resampled SDs (Appendix G); pairwise sign tests with explicit  $p$ -values; Bonferroni correction discussed where applicable.
- *Compute resources.* §5: total run cost  $\sim$  \$300–\$500 in commercial-API calls plus open-weights compute; protocol evaluable on a single A100 / RTX A6000 workstation.

## Reproducibility risks and mitigations.

- *API model deprecation.* §S (extended manifest): per-model risk classification (HuggingFace-pinned, API-snapshot, API-rolling); two open-weights models (Mistral-7B, Llama-3-8B) constitute the time-stable replication anchor.
- *Simulator stability.* The user simulator is itself an LLM; we use cross-family pairings to control for dual-role bias (§M, §N) and report a cross-simulator ablation showing  $\Delta\text{CBF} < 0.005$  across alternative simulator choices.
- *Bug-fix disclosure.* ✓ §A: prior versions (pre-fix) silently fell back algorithmic L2 baselines to LLM verbal estimates; the patch and full re-run are documented; pre-patch values are not used in any quantitative claim.
- *Adaptive policy R4 LOOCV gap.* ✓ R4 uses a fixed threshold  $\theta = 0$  by design (parameter-free); explicit LOOCV shows the gap from in-sample (0.722) to held-out (0.719) is only  $+0.003$ , confirming generalization (Appendix E).

## Ethics, broader impact, and licensing.

- *No human subjects in the released benchmark.* All “user” interactions are LLM-simulated personas; no human data collection. A planned future human-validation pilot study (50 sessions, 3 annotators) will require IRB approval.
- *Broader impact.* The benchmark exposes failure modes (LSA on GPT-5.5/Qwen-Plus; “feedback as friend or foe” split) directly relevant to deploying LLM agents that claim self-knowledge in production; we hope it informs deployment caution rather than enabling new harms.
- *Licensing.* LADDER released under MIT (code) and CC-BY 4.0 (data/traces), consistent with the Datasheet license. MMLU usage follows the original MMLU license.